where λ is the free-space wavelength, m is the diffraction order. Suppose that $n_s \approx 2.5$ (GaN), $n_a = 1$ (air), $\lambda = 500$ nm, $\Lambda = 300$ nm, find the critical angle and the first-order diffracted beams when $\theta_i = 60°$. What is your conclusion?

3.29 LED efficiencies A particular 890 nm IR LED for use in instrumentation has a AlGaAs chip. The active region has been doped p-type with 4×10^{17} cm^{-3} of acceptors and the nonradiative lifetime is about 60 ns. At a forward current of 50 mA, the voltage across it is 1.4 V, and the emitted optical power is 10 mW. Calculate the PCE, IQE, and EQE, and estimate the light extraction ratio. For AlGaAs, $B \approx 1 \times 10^{-16}$ m^3 s^{-1}.

3.30 LED luminous flux

(a) Consider a particular green LED based on InGaN MQW active region. The emission wavelength is 528 nm. At an LED current of 350 mA, the forward voltage is 3.4 V. The emitted luminous flux is 92 lm. Find the power conversion efficiency, external quantum efficiency, luminous efficacy, and the emitted optical power (radiant flux)? (Data for Osram LT CPDP.)

(b) A red LED emits 320 mW of optical power at 656 nm when the current is 400 mA and the forward voltage is 2.15 V. Calculate the power conversion efficiency, external efficiency and the luminous efficacy. (Data for thin film InGaAlP Osram LH W5AM LED.)

(c) A deep blue LED emits at an optical power of 710 mW at 455 nm when the current is 350 mA and the forward voltage is 3.2 V. Calculate the power conversion efficiency, external efficiency, and luminous efficacy. (Data for a GaN Osram LD W5AM LED.)

3.31 SLEDs and ELEDs Experiments carried out on an AlGaAs SLED (surface-emitting LED) and an ELED (edge-emitting LED) give the light output power vs. current data in Table 3.9. (a) Show that the output light power vs. current characteristics are not linear. (b) By plotting the optical power output (P_o) vs. current (I) data on a log-log plot show that $P_o \propto I^n$. Find n for each LED.

TABLE 3.9 Light output power vs. DC current for surface- and edge-emitting LEDs

SLED I (mA)	25	50	75	100	150	200	250	300
SLED Light output power, P_o (mW)	1.04	2.07	3.1	4.06	5.8	7.6	9.0	10.2
ELED I (mA)	25	50	75	100	150	200	250	300
ELED Light output power (mW) P_o	0.46	0.88	1.28	1.66	2.32	2.87	3.39	3.84

3.32 LED-Fiber coupling Efficiency

(a) It is found that approximately 200 μW is coupled into a multimode step index fiber from a surface-emitting LED when the current is 75 mA and the voltage across the LED is about 1.5 V. What is the overall efficiency of operation?

(b) Experiments are carried out on coupling light from a 1310 nm ELED (edge-emitting LED) into multimode and single-mode fibers. (i) At room temperature, when the ELED current is 120 mA, the voltage is 1.3 V and light power coupled into a 50 μm multimode fiber with NA (numerical aperture) of 0.2 is 48 μW. What is the overall efficiency? (ii) At room temperature, when the ELED current is 120 mA, the voltage is 1.3 V and light power coupled into a 9 μm single mode fiber is 7 μW. What is the overall efficiency?

3.33 Internal optical power For a particular AlGaAs LED emitting at 850 nm, the nonradiative recombination lifetime is $\tau_{nr} = 50$ ns. The recombination occurs in the active region, which has been doped with acceptors of concentration 10^{17} cm^{-3}, and the direct recombination coefficient B is 2×10^{-16} cm^3 s^{-1}. What is the internal optical power generated at a current of 100 mA?

3.34 Internal quantum efficiency (IQE) and bandwidth product Show that IQE and the bandwidth f_c product is

$$\eta_{\text{IQE}} f_c = \frac{1}{2\pi\tau_r}$$

What is your conclusion? Consider an LED in which the radiative and nonradiative lifetimes are approximately 10 ns and 50 ns, respectively. What are the IQE and the possible bandwidth f_c? What happens if defects cause τ_{nr} to be 25 ns? What other factors do you think would reduce the bandwidth?

3.35 Bandwidth and rise and fall time

(a) Suppose the rise τ_R for an LED is given, and the rise and fall times are the same. Show that its AC modulation bandwidth f_c (the cutoff frequency) is given by

$$f_c = 0.35/\tau_R$$

(b) The specifications for a particular 1300 nm InGaAs ELED (PE13W series) pigtailed to a fiber has rise and fall times given by 2.5 ns. Calculate the bandwidth f_c. How does that compare with the quoted value 150 MHz?

(c) At large currents, that is, when the minority injection is strong, the radiative recombination time τ_r is no longer constant, but given by Eq. (3.8.7), $\tau_r = 1/B\Delta n$, where Δn is the injected concentration of carriers. The injected carriers (Δn) into the active region are brought in by the current (J) and then they recombine and emit photons, that is, Δn is proportional to the current density J. How would this affect the bandwidth?

3.36 LED electronic drive circuits Figures 3.49 (b) shows one possible way to drive an LED. Assume that V_{CC} is large and that the base-emitter voltage of the BJT is 0.7 V, show that $I_F \approx V/R_F$. How large should V_{CC} be? Consider a red LED that has $V_F = 2\,\text{V}$ at $I_F = 20\,\text{mA}$ and $I_{\max} = 50\,\text{mA}$. If the input control voltage range is from 0 to 5V, what would you recommend for R_F, R_B, and V_{CC}?

3.37 LED electronic drive circuit design Figures 3.54 (a)–(c) shows three possible circuits that can be used to drive a high-current LED from CMOS or similar low-current logic circuits. A bipolar junction transistor (Q_1) is commonly used for this task.

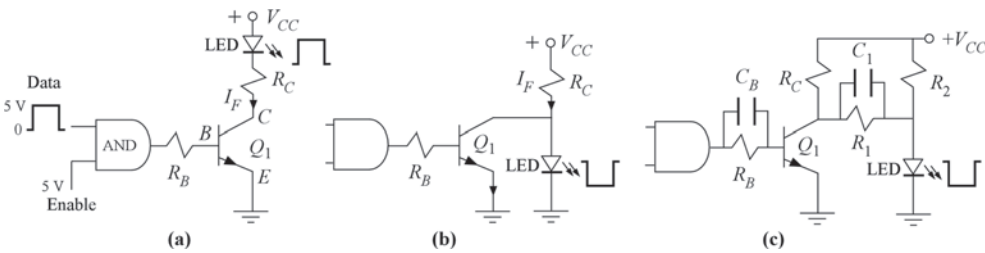
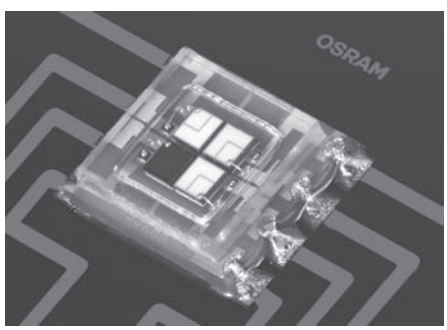


FIGURE 3.54 (a) A CMOS logic circuit (or a gate) drives an LED through a BJT Q_1. (b) The logic gate drives Q_1, which shunts the current from the LED and switches it off. (c) A combination of (a) and (b) with speeding capacitors C_B and C_1 to turn the LED on and off more quickly. In all three cases, the inputs to the logic circuit are the same.

(a) Consider Figure 3.54 (a), if the input is high (5 V), Q_1 is fully turned on and $V_{CE} = 0.25\,\text{V}$ (saturated CE voltage), and $V_{BE} = 0.7\,\text{V}$. What is the current I_F through the LED? If the LED needs $I_F = 50\,\text{mA}$ and $V_F = 2\,\text{V}$, and $V_{CC} = 5\text{V}$, what is R_C? Suppose that Q_1 has a minimum β (or h_{FE}) of 70, what is R_B? Is there an advantage in using higher V_{CC} voltages?

(b) Consider Figure 3.54 (b). Show that the LED is turned off when the input is high (5 V) [as opposed to (a) above]. For the same LED as in (a) with $I_F = 50\,\text{mA}$, $V_F = 2\,\text{V}$, and $V_{CC} = 5\text{V}$, find R_C and R_B.

(c) The circuit Figure 3.54 (c) has been *designed* to drive an LED that has $I_F = 25\,\text{mA}$ and $V_F = 2\,\text{V}$. The resistors are $R_B = 5.6\,\text{k}\Omega$, $R_C = 100\,\Omega$, $R_1 = 22\,\Omega$, and $R_2 = 820\,\Omega$. Explain how the circuit works, assuming first Q_1 is off and then it is on ($V_{CE} = 0.25\,\text{V}$). Is I_F close to 25 mA when Q_1 is off? The capacitors C_B and C_1 across R_B and R_1 speed up the circuit. What do you think C_BR_B and C_1R_1 should be? What are the requirement on Q_1?



This multichip LED from Osram is used in microprojectors. It is based on thin-film GaN technology. *(Courtesy of OSRAM Opto Semiconductors, Germany.)*



A handheld microprojector MPro120 from 3M for projecting photos and videos on a wall uses LEDs for its projection light. *(Courtesy of 3M.)*



LEDs have now widely replaced the incandescent light bulbs in traffic lights.



This LED-based light bulb for use in chandeliers uses a few watts of electric power (2.5W) to generate 135 lumens of light. There are 18 LED chips each with a yellow phosphor to generate the white light.

We thought it [the laser] might have some communications and scientific uses, but we had no application in mind. If we had, it might have hampered us and not worked out as well.

—Arthur Schawlow[1]



Zhores Alferov (on the right) with Valery Kuzmin (technician) in 1971 at the Ioffe Physical Technical Institute, discussing their experiments on heterostructures. Zhores Alferov carried out some of the early pioneering work on heterostructure semiconductor devices that led to the development of a number of important optoelectronic devices, including the heterostructure laser. Zhores Alferov and Herbert Kroemer shared the Nobel Prize in Physics (2000) with Jack Kilby. Their Nobel citation is "for developing semiconductor heterostructures used in high-speed- and opto-electronics." *(Courtesy of Professor Zhores Alferov, Ioffe Physical Technical Institute.)*

[1] Arthur Schawlow (1921–1999; Nobel Laureate, 1981) talking about the invention of the laser.

4 STIMULATED EMISSION DEVICES: OPTICAL AMPLIFIERS AND LASERS

4.1 STIMULATED EMISSION, PHOTON AMPLIFICATION, AND LASERS

A. Stimulated Emission and Population Inversion

An electron in an atom can be excited from an energy level E_1 to a higher energy level E_2 by the absorption of a photon of energy $h\upsilon = E_2 - E_1$ as shown in Figure 4.1 (a). When an electron at a higher energy level transits down in energy to an unoccupied energy level, it emits a photon. There are essentially two possibilities for the emission process. The electron can undergo the downward transition by itself quite *spontaneously*, or it can be *induced* to do so by another photon.

In **spontaneous emission**, the electron falls down in energy from level E_2 to E_1 and emits a photon of energy $h\upsilon = E_2 - E_1$ in a random direction as indicated in Figure 4.1 (b). The transition is spontaneous provided that the state with energy E_1 is empty, *i.e.,* not already occupied by another electron. In classical physics, when a charge accelerates and decelerates as in an oscillatory motion with a frequency υ it emits an electromagnetic radiation also of frequency υ. The emission process during the transition of the electron from E_2 to E_1 can be classically thought of as if the electron is oscillating with a frequency υ.

In **stimulated emission**[2], an incoming photon of energy $h\upsilon = E_2 - E_1$ stimulates the whole emission process by inducing the electron at E_2 to transit down to E_1. The emitted photon has the *same energy*, since $h\upsilon = E_2 - E_1$, it is *in phase* with the incoming photon, it is in the *same direction*, and it has the *same polarization*, that is, the optical field oscillations in the emitted and in the incoming photon are in the same direction, as illustrated in Figure 4.1 (c). To get a feel of what is happening during stimulated emission one can think of the electric field of the incoming photon coupling to the electron and thereby driving it with the same frequency as the incoming photon. The forced oscillation of the electron at a frequency $\upsilon = (E_2 - E_1)/h$ causes it to emit electromagnetic radiation whose electric field is in total phase with that of the stimulating radiation. When the incoming photon leaves the site, the electron has returned to E_1 because it has emitted a photon of energy $h\upsilon = E_2 - E_1$. Although we considered the

[2] Also called *induced emission.*

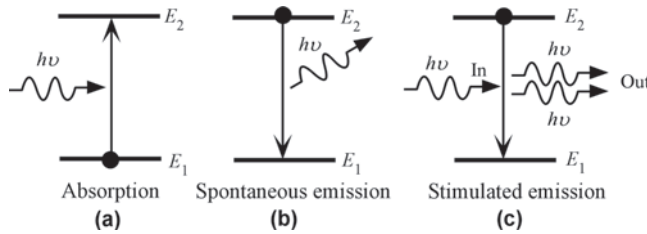


FIGURE 4.1 Absorption, spontaneous (random photon) emission, and stimulated emission.

transitions of an electron in an atom, we could have just well described photon absorption, and spontaneous and stimulated emission in Figure 4.1 in terms of energy transitions of the atom itself, in which case E_1 and E_2 represent the energy levels of the atom.

Stimulated emission is the basis for obtaining photon amplification since one incoming photon results in two outgoing photons which are in phase. How does one achieve a practical light amplifying device based on this phenomenon? It should be quite apparent from Figure 4.1 (c) that to obtain stimulated emission, the incoming photon should *not* be absorbed by another atom at E_1. When we are considering a collection of atoms to amplify light, we must therefore have the majority of the atoms at the energy level E_2. If this were not the case, the incoming photons would be absorbed by the atoms at E_1. When there are more atoms at E_2 than at E_1, we have what is called a **population inversion**. Under normal equilibrium conditions, as a result of Boltzmann statistics, most of the atoms would be at E_1, and very few at E_2. We therefore need to *excite* the atoms, cause population inversion, to obtain stimulated emission. It should be apparent that with *only two* energy levels, we can never achieve population at E_2 greater than that at E_1 because, in the steady state, the incoming photon flux will cause as many upward excitations as downward stimulated emissions. We need at least three energy levels as shown below in Section B.

B. Photon Amplification and Laser Principles

Let us consider the three-energy-level system shown in Figure 4.2 (a), with levels at E_1, E_2, and E_3. There are, of course, other energy levels but they are not involved in the following processes. The present example will roughly represent what happens in a ruby laser. The energy levels

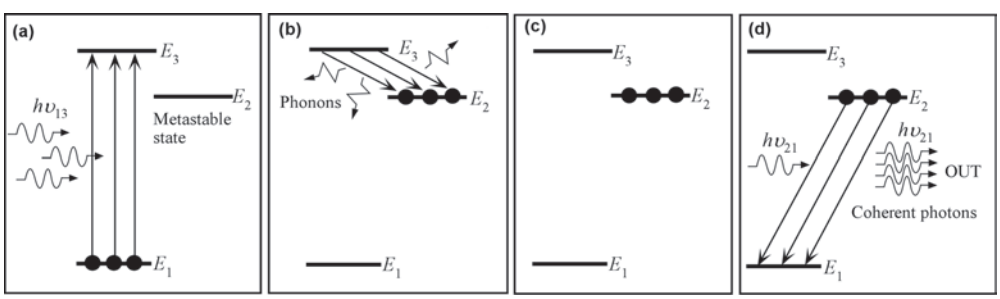


FIGURE 4.2 The principle of the laser, using a ruby laser as an example. (a) The ions (Cr^{3+}) in the ground state are pumped up to the energy level E_3 by photons from an optical excitation source. (b) Ions at E_3 rapidly decay to the long-lived state at the energy level E_2 by emitting lattice vibrations (phonons). (c) As the states at E_2 are long-lived, they quickly become populated and there is a population inversion between E_2 and E_1. (d) A random photon (from spontaneous decay) of energy $h\upsilon_{21} = E_2 - E_1$ can initiate stimulated emission. Photons from this stimulated emission can themselves further stimulate emissions, leading to an avalanche of stimulated emissions and hence coherent photons being emitted.

E_1, E_2, and E_3 correspond to the energies of the chromium ions (Cr^{3+}) in a crystal of Al_2O_3 (a corundum crystal); E_1 is the ground energy of the Cr^{3+} ion. Suppose that an external optical excitation causes the Cr^{3+} ions in the crystal to become excited to the energy level E_3. This is called the pump energy level, and the process of exciting the Cr^{3+} ions to E_3 is called **pumping**. In the present case, **optical pumping** is used, and we need a pump source with a photon energy of $h\upsilon_{13} = E_3 - E_1$.

From E_3, the Cr^{3+} ions decay rapidly to an energy level E_2 by emitting phonons,[3] that is, lattice vibrations (heat), as shown in Figure 4.2 (b). Thus, the transition from E_3 to E_2 is a radiationless transition. The energy level E_2 happens to correspond to a state that does not rapidly and spontaneously decay to the lower-energy state E_1. In other words, the state at E_2 is a **long-lived state**.[4] (The characteristic transition time constant from E_2 to E_1 is about 3 ms, which is very long in the atomic world.) Since the Cr^{3+} ions cannot decay rapidly from E_2 to E_1, they accumulate at this energy level. With ample pumping, we can accumulate sufficient Cr^{3+} ions at E_2 to cause a population inversion between E_2 and E_1, as pumping takes more and more Cr^{3+} ions to E_3 and hence E_2. Population inversion is shown in Figure 4.2 (c).

When a Cr^{3+} ion at E_2 decays spontaneously, it emits a photon (a "random photon") which can go on to a neighboring Cr^{3+} ion and cause that to execute stimulated emission. The photons from the latter can go on to the next Cr^{3+} ion at E_2 and cause that to emit by stimulated emission and so on. The result is an avalanche of stimulated emission processes with all the photons in phase, in the same direction, and with the same polarization, so that the light output is a large collection of *coherent photons* as in Figure 4.2 (d). This is the basic principle of the ruby laser, and hence the three-level laser. At the end of the avalanche of stimulated emission processes, the Cr^{3+} ions at E_2 would have dropped to E_1 and can be pumped again to repeat the stimulated emission cycle. The emission from E_2 to E_1 is called the **lasing emission**.[5] The system we have just described for photon amplification is a laser, an acronym for **Light Amplification by Stimulated Emission of Radiation**.[6]

A more realistic energy diagram for the Cr^{3+} ion in the ruby crystal is shown in Figure 4.3 (a), which is typical of three-level laser systems that involve ions in crystals or glasses. The main difference from the simple schematic in Figure 4.2 is that the higher energy level E_3 is actually a band of energy levels. There is another band of energies around E_3'. The Cr^{3+} ion can be excited by green light to E_3 and by blue light to E_3'. (The ruby crystal is "pink" because it absorbs green and blue light.[7]) Pumping is achieved by using an intense xenon flash light. The ions become excited to the E_3 and E_3' bands and then decay quickly from these bands by emitting phonons. They populate the E_2-level as described above. Once more than half the ions at E_1 have been pumped, there is population inversion between E_1 and E_2 and consequently the pumped medium (the ruby crystal) exhibits **optical gain**, or **photon amplification**.

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[3] A phonon is a quantum of lattice vibrational energy, just as a photon is a quantum of electromagnetic energy.

[4] We will not examine what causes certain states to be long-lived, but simply accept that these states do not decay rapidly by spontaneous emission to lower-energy states. Put differently, the decay time is very long (typically in milliseconds) on the atomic scale.

[5] *Arthur Schawlow*, one of the coinventors of the laser, was well-known for his humor and has, apparently, said that "Anything will lase if you hit it hard enough." In 1971, Schawlow and Ted Hänsch were able to develop the first edible laser made from Jell-O. (T. Hansch, M. Pernier, A. Schawlow, "Laser Action of Dyes in Gelation," *IEEE J. Quantum Elect.*, *7*, 45, 1971.)

[6] Although it is indeed an acronym, it is now commonly used as a noun that implies a device that emits coherent radiation.

[7] The ruby crystal appears red if the Cr^{3+} ions concentration is large. For the Cr^{3+} concentrations used in the ruby laser, the crystal appears pink.

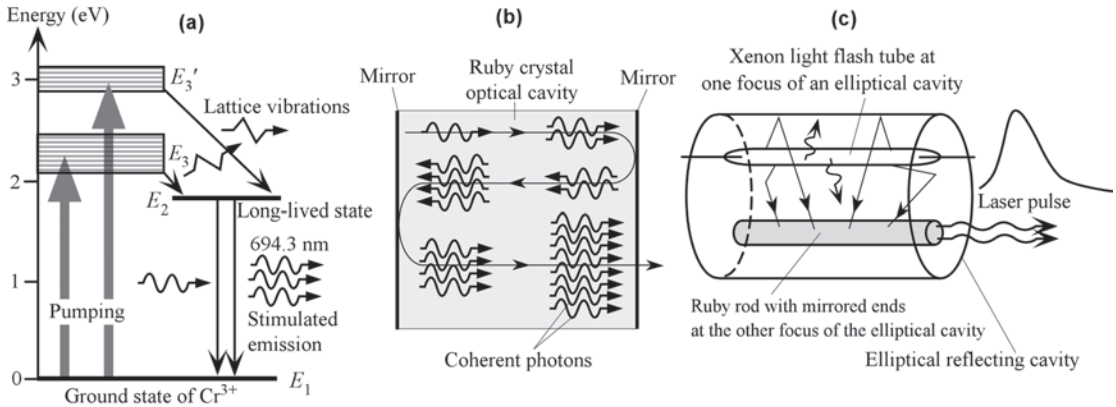


FIGURE 4.3 (a) A more realistic energy diagram for the Cr^{3+} ion in the ruby crystal (Al_2O_3), showing the optical pumping levels and the stimulated emission. (b) The laser action needs an optical cavity to reflect the stimulated radiation back and forth to build up the total radiation within the cavity, which encourages further stimulated emissions. (c) A typical construction for a ruby laser, which uses an elliptical reflector, and has the ruby crystal at one focus and the pump light at the other focus.

To get coherent lasing radiation out from the ruby laser, we need to do more, not just amplify the radiation along the ruby crystal. We can increase stimulated emissions by increasing the number of photons, *i.e.,* the radiation intensity, within the crystal inasmuch as more photons cause more stimulated emissions. The ends of the ruby crystal, which is normally a rod, are silvered to reflect back and forward the stimulated radiation, that is, to form an **optical cavity** with mirrors at the ends, as shown Figure 4.3 (b). As the stimulated photons are reflected back into the crystal, the radiation intensity buildsup inside the crystal, in much the same way we build up voltage oscillations in an electrical oscillator circuit by feedback. The buildup of coherent radiation in the cavity encourages further stimulated emissions, until a large avalanche of stimulated transitions occur and takes most of the ions at E_2 down to E_1. One of the mirrors is partially silvered to allow some of this radiation to be tapped out. What comes out is a pulse of highly coherent radiation that has a high intensity as illustrated in Figure 4.3 (b).

The coherency and the well-defined wavelength of the emitted radiation from a laser are attributes that make it distinctly different from a random stream of different wavelength photons emitted from a tungsten bulb, or randomly phased photons from an LED. The reader might have noticed that the photon energy emitted from the laser system is less than the photon energy we used to pump it, that is, excite it; $h\nu_{21} < h\nu_{13}$. However, we only needed incoherent radiation to pump the system, and we obtained a fully coherent radiation as output.

Practical ruby lasers need efficient optical pumping, which can be obtained by using an elliptical reflector with the ruby crystal rod at one focus, and the pump light, a xenon flash, at the other focus as shown in Figure 4.3 (c). The early ruby lasers used a helical xenon flash tube surrounding the ruby rod. The lasing emission from the ruby laser is a light pulse, whose duration and intensity depend on the laser construction, and the xenon flash. Ruby lasers are frequently used in interferometry, holography, and hair and tattoo removal, among other applications.





Theodore Harold Maiman was born in 1927 in Los Angeles, son of an electrical engineer. He studied engineering physics at Colorado University, while repairing electrical appliances to pay for college, and then obtained a Ph.D. from Stanford. Theodore Maiman constructed this first laser in 1960 while working at Hughes Research Laboratories (T.H. Maiman, "Stimulated optical radiation in ruby lasers", *Nature*, **187**, 493, 1960). There is a vertical chromium ion doped ruby rod in the center of a helical xenon flash tube. The ruby rod has mirrored ends. The xenon flash provides optical pumping of the chromium ions in the ruby rod. The output is a pulse of red laser light. *(Courtesy of HRL Laboratories, LLC, Malibu, California.)*

C. Four-Level Laser System

The problem with the three-level laser system in Figure 4.2 (a) is that to achieve population inversion, we must pump at least half the Cr^{3+} ions at the ground level E_1 to E_2. It would be much better if we could achieve population inversion as soon as we begin exciting the ions at E_1 to E_3 to E_2. We can quickly and easily achieve population inversion if we use a four-level system as in Figure 4.4, where the ground energy is E_0. Figure 4.4 roughly represents the energy levels of Nd^{3+} ions in a $Y_3Al_5O_{11}$ (yttrium aluminate garnate, YAG) crystal, usually written as Nd^{3+}: YAG. In equilibrium, the population of the ions at these energy levels follow the Boltzmann distribution; and nearly all the ions are at E_0 whereas E_1, E_2, and E_3 are mostly empty.

The ions at E_0 are excited by optical pumping to the energy level E_3. In many cases, this is a band of energy levels and is called the **pump band**. From E_3, the ions decay rapidly by phonon emission to E_2, which is a long-lived state with a lifetime in the milliseconds, *i.e.,* the spontaneous decay from E_2 to E_1 is slow. Clearly, as soon as ions begin to populate E_2 we have population inversion between the level E_1 and E_2 inasmuch as E_1 is initially nearly empty. The lasing emission takes place by stimulated emission as ions drop from E_2 to E_1. From E_1, the ions return back to E_0 by the emission of phonons (or spontaneous photons in some cases). We also need the ions at E_1 to decay quickly back to E_0 so that a buildup of ions at E_1 is avoided, and the ions are quickly returned back to E_0 for repumping. In cases where E_1 is within only a few k_BT to E_0, the four-level system behaves almost like the three-level system in Figure 4.2.

The four-level laser system in Figure 4.4 serves as an excellent way to explain the majority of practical lasers. Indeed, most modern lasers are based on a four-level system, or its equivalent. For example, in the Nd^{3+}: YAG laser, the Nd^{3+} ions are excited by using a flash tube or a laser diode (LD) to E_3 and hence to E_2, which is the *upper* laser level. The lifetime of the ions at E_2 is about 230 μs. The *lower* laser level is E_1. The lasing emission from E_2 to E_1 is in the infrared at 1064 nm. The Nd^{3+}:YAG can be operated in the pulsed mode as well as in CW mode. In contrast, the ruby laser is operated in the pulsed mode because of the large power required for pumping the ruby crystal.

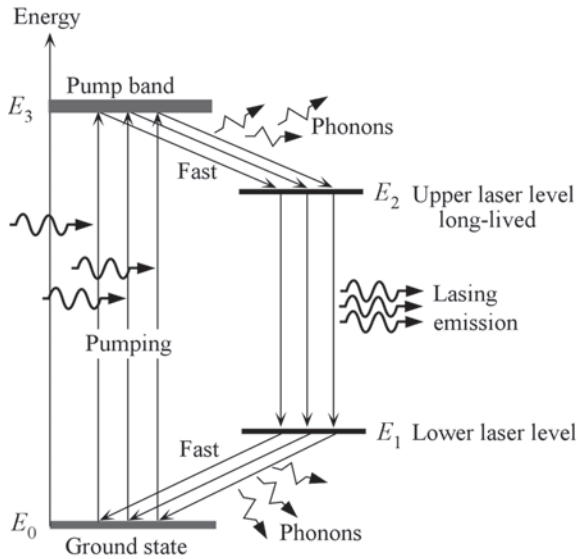


FIGURE 4.4 A four-energy-level laser system. Highly simplified representation of Nd^{3+}:YAG laser.

4.2 STIMULATED EMISSION RATE AND EMISSION CROSS-SECTION

A. Stimulated Emission and Einstein Coefficients

A useful laser medium must have a higher efficiency of stimulated emission compared with the efficiencies of spontaneous emission and absorption. We need to determine the controlling factors for the rates of stimulated emission, spontaneous emission, and absorption. Consider a medium as in Figure 4.5 that has N_1 atoms per unit volume at the energy E_1 and N_2 atoms per unit volume at the energy E_2. Then the rate of upward transitions from E_1 to E_2 by photon absorption will be proportional to the concentration of atoms N_1, and also to the number of photons per unit volume with energy $h\upsilon = E_2 - E_1$. Put differently, this rate will depend on the energy density in the radiation. Thus, the upward transition rate R_{12} is

Upward transitions

$$R_{12} = B_{12}N_1\rho(\upsilon) \tag{4.2.1}$$

where B_{12} is a proportionality constant termed the **Einstein B_{12} coefficient**, and $\rho(\upsilon)$ is the photon energy density per unit frequency,[8] that is, the energy in the radiation per unit volume

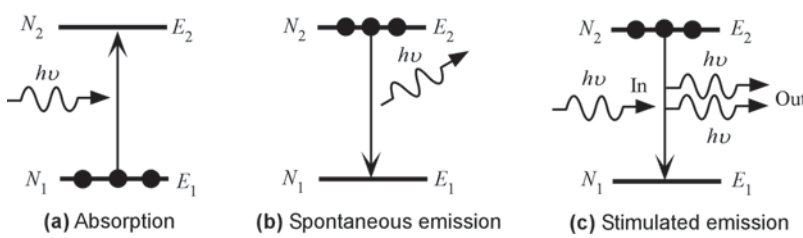


FIGURE 4.5 Absorption, spontaneous emission, and stimulated emission.

[8]Using $\rho(\upsilon)$ defined in this way simplifies the evaluation of the proportionality constants in this section. Further, in the simplified treatise here, we neglected the degeneracy of the energy levels, which introduces numerical factors that multiply N_1 and N_2.

per unit frequency due to photons with energy $h\upsilon = E_2 - E_1$. The rate of downward transitions from E_2 to E_1 involves spontaneous and stimulated emission. The spontaneous emission rate depends on the concentration N_2 of atoms at E_2. The stimulated emission rate depends on both N_2 and the photon concentration $\rho(\upsilon)$ with energy $h\upsilon (= E_2 - E_1)$. Thus, the total downward transition rate is

$$R_{21} = A_{21}N_2 + B_{21}N_2\rho(\upsilon) \tag{4.2.2}$$

Downward transitions

where the first term is due to spontaneous emission [does not depend on the photon energy density $\rho(\upsilon)$ to drive it] and the second term is due to stimulated emission, which requires photons to drive it. A_{21} and B_{21} are the proportionality constants termed the **Einstein coefficients** for spontaneous and stimulated emissions, respectively.

To find the coefficients A_{21}, B_{12}, and B_{21}, we consider the events in equilibrium, that is, the medium in thermal equilibrium (no external excitation). There is no net change with time in the populations at E_1 and E_2 which means

$$R_{12} = R_{21} \tag{4.2.3}$$

Furthermore, in thermal equilibrium, Boltzmann statistics demands that

$$N_2/N_1 = \exp\left[-(E_2 - E_1)/k_BT\right] \tag{4.2.4}$$

Boltzmann statistics in equilibrium

where k_B is the Boltzmann constant and T is the absolute temperature.

Now in *thermal equilibrium*, in the collection of atoms we are considering, radiation from the atoms must give rise to an equilibrium photon energy density, $\rho_{\text{eq}}(\upsilon)$, that is given by *Planck's black body radiation distribution law*[9]

$$\rho_{\text{eq}}(\upsilon) = \frac{8\pi h\upsilon^3}{c^3\left[\exp\left(\dfrac{h\upsilon}{k_BT}\right) - 1\right]} \tag{4.2.5}$$

Planck's black body radiation in equilibrium

It is important to emphasize that Planck's law in Eq. (4.2.5) applies only in thermal equilibrium; we are using this equilibrium condition to determine the Einstein coefficients. During the laser operation, of course, $\rho(\upsilon)$ is not described by Eq. (4.2.5); in fact it is much larger. From Eqs. (4.2.1) to (4.2.5) we can readily show that

$$B_{12} = B_{21} \tag{4.2.6}$$

Absorption and stimulated emission coefficients

and

$$A_{21}/B_{21} = 8\pi h\upsilon^3/c^3 \tag{4.2.7}$$

Spontaneous and stimulated emission coefficients

Now consider the ratio of stimulated to spontaneous emission

$$\frac{R_{21}(\text{stim})}{R_{21}(\text{spon})} = \frac{B_{21}N_2\rho(\upsilon)}{A_{21}N_2} = \frac{B_{21}\rho(\upsilon)}{A_{21}} \tag{4.2.8}$$

[9] See, for example, any modern physics textbook.

which, by Eq. (4.2.7), can be written as

Stimulated to spontaneous emission

$$\frac{R_{21}(\text{stim})}{R_{21}(\text{spon})} = \frac{c^3}{8\pi h\upsilon^3}\rho(\upsilon) \tag{4.2.9}$$

In addition, the ratio of stimulated emission to absorption is

Stimulated to absorption

$$\frac{R_{21}(\text{stim})}{R_{12}(\text{absorp})} = \frac{N_2}{N_1} \tag{4.2.10}$$

There are two important conclusions. For stimulated photon emission to exceed photon absorption, by Eq. (4.2.10), we need to achieve **population inversion**, that is, $N_2 > N_1$. For stimulated emission to far exceed spontaneous emission, by Eq. (4.2.9) we must have a large photon concentration, which is achieved by building an **optical cavity** to contain the photons.

The rate R_{12} represents the rate at which N_1 is decreasing by absorption, thus, $R_{12} = -dN_1/dt$. The rate R_{21} represents the rate at which N_2 is decreasing by spontaneous and stimulated emission, thus, $R_{21} = -dN_2/dt$. Further, the rate at which N_2 changes by spontaneous emission can be written as

Spontaneous decay lifetime

$$dN_2/dt = -A_{21}N_2 = -N_2/\tau_{\text{sp}} \tag{4.2.11}$$

where, by definition, $\tau_{\text{sp}} = 1/A_{21}$, and is termed the **spontaneous decay time**, or more commonly called the **lifetime** of level E_2.

It is important to point out that the population inversion requirement $N_2 > N_1$ means that we depart from thermal equilibrium. According to Boltzmann statistics in Eq. (4.2.4), $N_2 > N_1$ implies a *negative temperature*! The laser principle is based on **non-thermal equilibrium**.[10]

The number of states an ion or an atom has at a given energy level is called its **degeneracy** g. We assumed that the degeneracy g_1 and g_2 of E_1 and E_2 are the same. We can, of course, redo the treatment above by including g_1 and g_2. The final result is equivalent to replacing N_1 by $N_1(g_2/g_1)$.

EXAMPLE 4.2.1 Minimum pumping power for three-level laser systems

Consider the three-level system as shown in Figure 4.2 (a). Assuming that the transitions from E_3 to E_2 are fast, and the spontaneous decay time from E_2 to E_1 is τ_{sp}, show that the *minimum* pumping power $P_{p\min}$ that must be absorbed by the laser medium per unit volume for population inversion ($N_2 > N_1$) is

Minimum pumping for population inversion for three-level laser

$$P_{p\min}/V = (N_0/2)h\upsilon_{13}/\tau_{\text{sp}} \tag{4.2.12}$$

where V is the volume, N_0 is the concentration of ions in the medium, and hence at E_0 before pumping. Consider a ruby laser in which the concentration of Cr^{3+} ions is 10^{19} cm^{-3}, the ruby crystal rod is 10 cm long and 1 cm in diameter. The lifetime of Cr^{3+} at E_2 is 3 ms. Assume the pump takes the Cr^{3+} ions to the E_3-band in Figure 4.3 (a), which is about 2.2 eV above E_0. Estimate the minimum power that must be provided to this ruby laser to achieve population inversion.

[10] "But I thought, now wait a minute! The second law of thermodynamics assumes thermal equilibrium. We don't have that!" *Charles D. Townes* (born 1915; Nobel Laureate, 1964). The laser idea occurred to Charles Townes, apparently, while he was taking a walk one early morning in Franklin Park in Washington, DC, while attending a scientific committee meeting. Non-thermal equilibrium (population inversion) is critical to the principle of the laser.

Solution

Consider the three-level system in Figure 4.2 (a). To achieve population inversion we need to get half the ions at E_1 to level E_2 so that $N_2 = N_1 = N_0/2$, since N_0 is the total concentration of Cr^{3+} ions all initially at E_1. We will need $\left[(N_0/2)h\upsilon_{13} \times \text{volume}\right]$ amount of energy to pump to the E_3-band. The ions decay quickly from E_3 to E_2. We must provide this pump energy before the ions decay from E_2 to E_1, that is, before τ_{sp}. Thus, the *minimum* power the ruby needs to absorb is

$$P_{p\min} = V(N_0/2)h\upsilon_{13}/\tau_{sp}$$

which is Eq. (4.2.12). For the ruby laser

$$P_{p\min} = \left[\pi(0.5\text{ cm})^2(10\text{ cm})\right]\left[(10^{19}\text{ cm}^{-3})/2\right]\left[(2.2\text{ eV})(1.6 \times 10^{-19}\text{ J/eV})\right]/(0.003\text{ s}) = 4.6\text{ kW}$$

The total pump energy that must be provided in less than 3 ms is 13.8 J.

B. Emission and Absorption Cross-Sections

In many photonic applications, we need to describe absorption and emission in terms of cross-sections. Consider a medium that has been doped with ions, perhaps Nd^{3+} ions in YAG or glass, and consider two energy levels E_1 and E_2, as in Figure 4.6 (a). Suppose that there is a flux of photons Φ_{ph} passing through the medium along a particular direction (x), and the photon energy $h\upsilon = E_2 - E_1$. The flux is the number of photons flowing per unit area along a certain direction so that $\Phi_{ph} = I/h\upsilon$, where I is the intensity of the beam[11] (radiation energy flow per unit area per unit time). The absorption of radiation, that is, photons, in the medium depends on two factors. First is the number of ions per unit volume, N_1; higher concentration N_1 will mean more absorption. Second is the nature of the interaction of the photon with an individual ion, and is related to the properties of the ion in the medium. The interaction of a photon with an individual ion can be intuitively visualized as the ion having a certain *cross-sectional area*, σ_{ab}, and if the photon

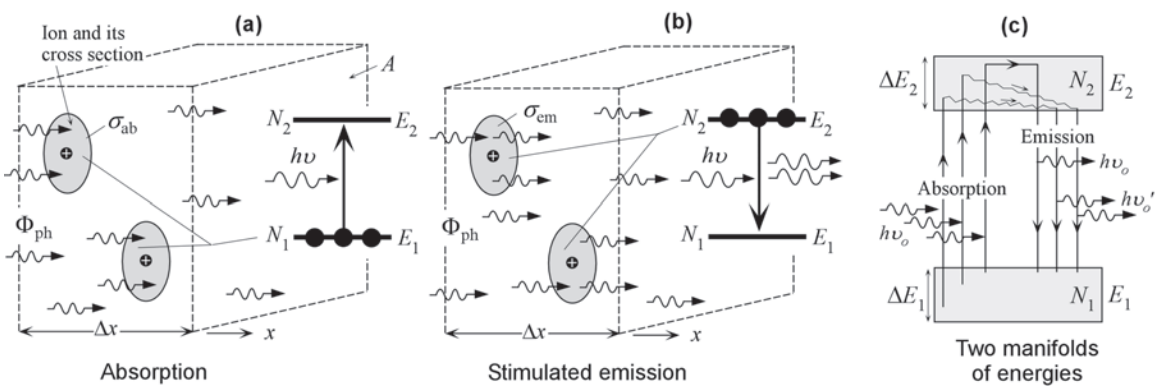


FIGURE 4.6 (a) Absorption of photons by ions in a medium, and the illustration of the absorption cross-section σ_{ab}. (b) Stimulated emission of photons by the same ions in the medium, and the illustration of the emission cross-section σ_{em}. (c) Upper and lower manifolds of energy. Each manifold has numerous closely spaced sublevels. Absorption and emission can involve the same or different levels.

[11] I (san-serif font) is used for the light intensity (optical power flow per unit area, W m^{-2}) and I for the current in this chapter.

crosses this area, it becomes absorbed, as illustrated in Figure 4.6 (a). The optical power absorbed by a single ion depends on the light intensity incident *and* the ion's absorption cross-section, *i.e.,*

Definition of absorption cross-section

$$\text{Optical power absorbed by an ion} = \text{Light intensity} \times \text{Absorption cross-section of ion} = I\sigma_{ab} \quad \textbf{(4.2.13)}$$

which defines the **absorption cross-section** σ_{ab}. For the collection of N_1 ions per unit volume, the total power absorbed per unit volume is $I\sigma_{ab}N_1$.

Consider a small volume $A\Delta x$ of the medium, as in Figure 4.6 (a), where A is the cross-sectional area and the radiation propagates along x. The total optical power ΔP_o absorbed in this volume is then $(I\sigma_{ab}N_1)(A\Delta x)$. Suppose that ΔI is the *change* (negative if absorption) in the intensity of the propagating radiation over the distance Δx. Then

$$A\Delta I = -\text{Total power absorbed in } A\Delta x = -(I\sigma_{ab}N_1)(A\Delta x)$$

i.e.,

Absorption cross-section σ_{ab} and coefficient α

$$-\frac{\Delta I}{I\Delta x} = \sigma_{ab}N_1 = \alpha \quad \textbf{(4.2.14)}$$

where the left-hand side is the fractional change in the light intensity per unit distance, and therefore represents α, the **absorption coefficient** of the medium, due to the absorption of radiation by ions at E_1, which reach E_2. Clearly, $\alpha = \sigma_{ab}N_1$. We could have easily used the latter as the definition of σ_{ab} and derived Eq. (4.2.13) instead.

The stimulated emission case, shown in Figure 4.6 (b), is similar. We can define an **emission cross-section** σ_{em} such that if an incident photon crosses the area σ_{em}, it stimulates the ion to emit a photon as shown in Figure 4.6 (b). In a similar way to Eq. (4.2.13)

Definition of emission cross-section

$$\text{Stimulated optical power emitted by an ion} = \text{Light intensity} \times \text{Emission cross-section of ion} = I\sigma_{em} \quad \textbf{(4.2.15)}$$

As the radiation propagates along x, its intensity I increases due to stimulated emissions from E_2 to E_1. Since only ions at E_2 with a concentration N_2 can be stimulated to emit, the equivalent of Eq. (4.2.14) for the fractional increase in the intensity per unit distance is

Emission cross-section σ_{am} and optical gain

$$\frac{\Delta I}{I\Delta x} = \sigma_{em}N_2 \quad \textbf{(4.2.16)}$$

The net fractional increase in the intensity I per unit distance is defined as the **gain coefficient** g of the medium, so that, in the presence of stimulated emission and absorption

Net optical gain coefficient

$$g = \left[\frac{\Delta I}{I\Delta x}\right]_{net} = \sigma_{em}N_2 - \sigma_{ab}N_1 \quad \textbf{(4.2.17)}$$

In the presence of just two discrete energy levels, as in Figure 4.6 (a) and (b), we know that $B_{12} = B_{21}$, and similarly, $\sigma_{ab} = \sigma_{em}$.

However, in many practical cases, the two energy levels are actually two manifolds of energies, that is, each manifold consists of narrowly spaced sublevels, as illustrated in Figure 4.6 (c). The lower E_1-manifold has sublevels that cover the range ΔE_1, as in Figure 4.6 (c), and the higher E_2-manifold has sublevels that are within ΔE_2. Suppose we excite the ions with photons of energy $h\nu_o$ that are absorbed and take some of ions from the E_1- to the E_2-manifold. Three of the

possibilities for absorption are shown in Figure 4.6 (c) that take ions from different sublevels in ΔE_1 and put them at various sublevels in ΔE_2 in which $h\upsilon_o$ is the same for all. However, the downward transitions are more selective because the end level in the E_1-manifold must be empty, and only the higher levels in this manifold (top of ΔE_1) are less populated. Thus, there are less chances for the downward transitions that emit the same $h\upsilon_o$ as shown in Figure 4.6 (c). In addition, the excited ions in the E_2-manifold decay rapidly by emitting lattice vibrations (they thermalize) to the lower levels in E_2 and then emit photons with $h\upsilon_o' < h\upsilon_o$. Clearly, the net effect is that emission at υ_o is less likely than absorption at the same frequency. Consequently, σ_{em} and σ_{ab} are then *not* the same at υ_o; and σ_{ab} is larger than σ_{em} in this example.

We can consider other photon energies besides $h\upsilon_o$ in Figure 4.6 (c), and hence consider σ_{ab} and σ_{em} at other frequencies. We must therefore generalize Eq. (4.2.17) and write σ_{ab} and σ_{em} as frequency dependent, *i.e.,*

$$g(\upsilon) = \sigma_{\text{em}}(\upsilon)N_2 - \sigma_{\text{ab}}(\upsilon)N_1 \tag{4.2.18}$$

Optical gain coefficient

Although Eq. (4.2.18) has been derived in terms of υ as a variable, it could have just as well be written in terms of λ.

When the net gain in Eq. (4.2.18) is zero, the medium becomes **transparent** to the incoming radiation, which is neither absorbed nor amplified. The **transparency condition** is

$$\sigma_{\text{em}}(\upsilon)N_2 - \sigma_{\text{ab}}(\upsilon)N_1 = 0 \tag{4.2.19}$$

Transparency condition

To find the overall optical power (or intensity) gain G through a medium that has been pumped or excited, we need to integrate Eq. (4.2.17) along the length of the medium, from $x = 0$ to $x = L$. If N_2 and N_1 remain the same over L and hence g is constant along L, then $G = I(L)/I(0)$ is given by

$$G = \exp(gL) \tag{4.2.20}$$

Net optical power gain

Henceforth we will use the above equations in describing optical amplifiers.

EXAMPLE 4.2.2 Gain coefficient in a Nd^{3+}-doped glass fiber

Consider a Nd^{3+}-doped silica-based glass fiber. Suppose we want to use this fiber as an optical amplifier. The Nd^{3+} concentration in the fiber core is $1 \times 10^{19}\,\text{cm}^{-3}$. The Nd^{3+} ions can be pumped from E_0 to E_3 in Figure 4.4 from which they decay rapidly down to E_2, and the population inversion between E_2 and E_1 gives optical amplification. The E_2 to E_1 emission is approximately at 1.05 μm, and the emission cross-section is about $3 \times 10^{-20}\,\text{cm}^2$. Estimate the maximum gain coefficient with this amount of doping if we could pump all the Nd^{3+} ions. What is gain G in dB if the fiber is 20 cm long?

Solution

The maximum gain will be achieved when all Nd^{3+} ions have been pumped to E_2. From Eq. (4.2.17)

$$g = \sigma_{\text{em}}N_2 = (3 \times 10^{-20}\,\text{cm}^2)(1 \times 10^{19}\,\text{cm}^{-3}) = 0.3\,\text{cm}^{-1} \quad \text{or} \quad 30\,\text{m}^{-1}$$

This gain increases with the Nd^{3+} concentration. The maximum theoretical gain G_{max} for a 20 cm fiber is

$$G = \exp(gL) = \exp\left[(0.3\,\text{cm}^{-1}) \times (20\,\text{cm})\right] = 403.4 \quad \text{or} \quad 26.1\,\text{dB}$$

only with those atoms that have their individual gain curves overlapping the υ_1-radiation. When the υ_0-radiation de-excites atoms, it does not de-excite those that are at frequencies υ_1 and υ_1'. We know that stimulated emissions de-excite atoms and reduce the population inversion and hence the gain. Consequently, under steady state operation, the gain for the υ_0-mode is reduced from its peak g_0 down to g_{th}, which is the required gain for maintaining steady state oscillations. Similarly, the gain for υ_1-mode is reduced from g_1 down to g_{th}, as shown in Figure 4.27 (e). To maintain steady state laser oscillations, $g(\upsilon)$ must have dipped regions at υ_0, υ_1, and υ_1' where $g(\upsilon)$ becomes g_{th}. The dip in $g(\upsilon)$ at a mode is called **spectral hole burning**. It might seem as though all three modes should have the same intensity as they all have $g = g_{th}$, but this is not the case. Consider the υ_0- and υ_1-modes. The reduction of the gain from its peak at g_0 to g_{th}, and its maintenance at g_{th} involves stimulated emissions for de-excitation. More stimulated emissions are needed to reduce g_0 to g_{th} than for reducing g_1 to g_{th}. Thus υ_0-mode has greater stimulated emissions and more optical power than the υ_1-mode. The output spectrum is illustrated in Figure 4.27 (f). (As one might have surmised, for simplicity only, the mode υ_0 was made coincident with the peak of the gain curve.)

4.8 PULSED LASERS: *Q*-SWITCHING AND MODE LOCKING

A. *Q*-Switching

The *Q*-factor of an optical resonant cavity, like its finesse, is a measure of the cavity's frequency selectiveness. The higher the *Q*-factor, the more selective the resonator is, or the narrower the spectral width. It is also a measure of the energy stored in the resonator per unit energy dissipated, *e.g.,* losses at the reflecting surfaces or scattering in the cavity, per cycle of oscillation.[27] ***Q*-switching** refers to a laser whose optical resonant cavity is switched from a low *Q* to a high *Q* to generate an intense laser pulse, as described below.

A laser operation needs a good optical resonant cavity with low losses, that is, a high *Q*-factor, to allow the gain to reach the threshold gain. The large radiation intensity within the cavity stimulates further emissions. However, by temporarily removing one of the reflectors as in Figure 4.28 (a), we can switch the *Q*-factor of the cavity to some low value. While the optical

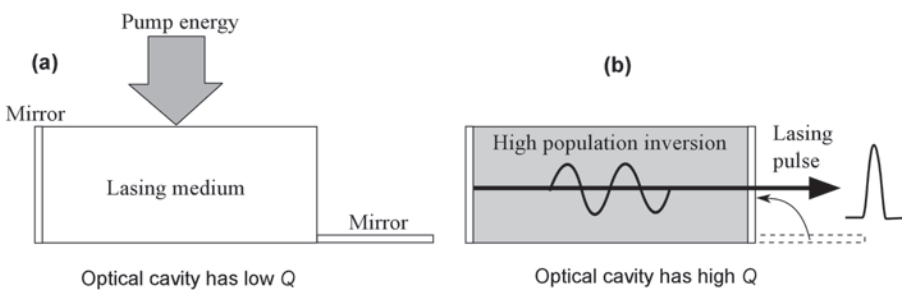


FIGURE 4.28 (a) The optical cavity has a low *Q* so that pumping takes the atoms to a very high degree of population inversion; lasing is prevented by not having a right-hand mirror. (b) The right mirror is "flung" to make an optical resonator, *Q* is switched to a high value which immediately encourages lasing emissions. There is an intense pulse of lasing emission which brings down the excess population inversion.

[27] See Section 1.11 in Chapter 1. All optical resonators have a *Q*-factor, which is another way of expressing their finesse.

cavity has low Q, the lasing is suppressed. Without the reflections from the mirrors, the photon energy density in the cavity is small, and the active medium can be pumped to achieve a large population inversion, greater than what would have normally been achieved had the mirrors been in place. When the Q of the optical cavity is switched to a high value, as in Figure 4.28 (b), an intense lasing emission is generated. While the Q is low, the pumped lasing medium is effectively a very high gain photon amplifier. There is too much loss in the optical resonator (*i.e.*, no optical feedback) to achieve a lasing oscillation. As soon as the Q is switched to a high value, the low loss in the optical resonator allows lasing oscillations to occur, which deplete the population inversion and decrease the gain until the population inversion falls below the threshold value and lasing oscillations cease.

The sequence of events involved in Q-switching a laser is illustrated in Figure 4.29. The pump, a flash tube in many cases, that provides optical excitation, is triggered at time t_1 as shown in Figure 4.29 (a). It must have sufficient intensity to quickly build up (N_2-N_1) to well above the normal threshold value $(N_2 - N_1)_{\text{th}}$ as indicated in Figure 4.29 (b). The pumping power needed must be provided before excited ions (or atoms) in the medium can spontaneously decay to the lower laser energy level. The optical cavity's Q is switched at time t_2, before the excited ions decay spontaneously to the lower laser level, as shown in Figure 4.29 (c). The result is a nearly sudden formation of an optical cavity, which enables lasing radiation to be quickly built up inside the cavity from t_2 to t_3 in Figures 4.29 (c)–(d). The radiation buildup from t_2 to t_3 is of the order of nanoseconds and starts with stray photons as in normal CW laser operation. Eventually, the rapid build-up of radiation in the cavity stimulates lasing transitions, and acts as positive feedback, yielding a pulse of coherent radiation as output as in Figure 4.29 (d). The duration of the lasing emission (the width of the output pulse) depends on both the Q-switching time, and the photon cavity lifetime, *i.e.*, how long photons are in the cavity before they leave. The Q-switching speed must be faster than the rate of oscillation buildup; otherwise the lasing oscillation builds up slowly. Table 4.3 summarizes the properties of two commercial Q-switched Nd^{3+}:YAG lasers. Notice that the Q-switched pulses are a few

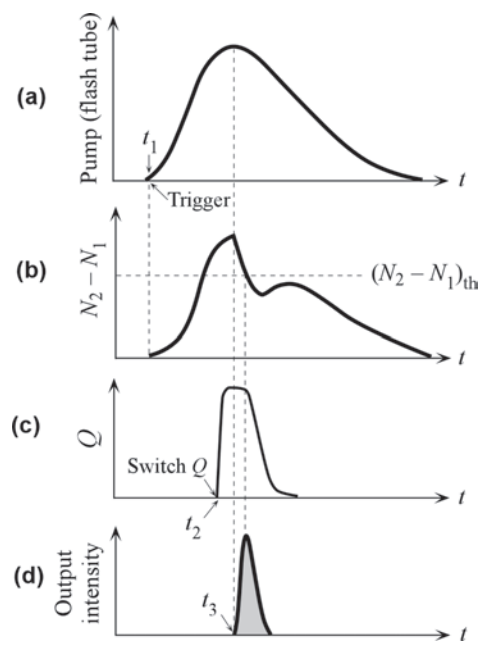


FIGURE 4.29 A simplified schematic of Q-switching in the generation of short laser pulses. (a) The pump profile, *i.e.,* the flash tube output. (b) The evolution of the population difference $N_2 - N_1$ with time. (c) The switching of the optical cavity Q during the pumping procedure while the population inversion is very large and greater than the normal threshold population. (d) The output light pulse.

TABLE 4.3 Selected typical characteristics of two commercial *Q*-switched Nd^{3+}:YAG lasers

Property	Nd^{3+}:YAG NL220 by EKSPLA	Nd^{3+}:YAG P-1064-150-HE by Alphalas
Emission wavelength (nm)	1064	1064
Output pulse energy (mJ)	10	1.5
Pulse width (ns)	7	1.1
Repetition rate (Hz)	1000	100
Peak power (kW)	~1400	~1400
Q-switch	EO	Passive
Output beam	TEM_{00}	TEM_{00}
Output beam M^2	<1.5	–
Beam divergence (mrad)	<1.5	6
Beam diameter (mm)	2.5	0.3

Note: Usually higher-energy output pulses require operation at a lower repetition frequency.

nanoseconds long, and the actual power in the laser pulse is 1.4–1.8 MW; extremely high compared with CW power levels.

There are a number of ways in which *Q*-switching can be conveniently implemented as illustrated in three examples in Figures 4.30 (a)–(c). The first example uses a rotating reflector, which in this case is a prism. The cavity *Q* is switched during the rotation of the prism every time a prism face aligns with the cavity axis, which is shown in Figure 4.30 (a). Another technique is to use a *saturable absorber*, also called a *bleachable dye*, in the optical cavity as in Figure 4.30 (b). A saturable absorber's absorption *decreases* with *increasing* light intensity so that it becomes transparent only at high intensities. When the absorber becomes transparent under a large radiation intensity, the dye is said to be bleached. Initially the absorber is opaque and keeps the cavity *Q* low by absorbing the radiation, but as the intensity builds up the absorber becomes transparent, which corresponds to switching the cavity *Q* to a high value. Another technique that is widely employed is the use of an *electro-optic* (EO) *switch* in the optical cavity as illustrated in Figure 4.30 (c). As will be explained in Chapter 6, such an EO switch is essentially an electro-optic (or a piezoelectric) crystal with electrodes, and works in combination with a polarizer. EO switches can be very fast (*e.g.*, less than nanoseconds), which is one of their distinct advantages. For the present purposes it is clear that when the EO switch is turned transparent by the application of a voltage, the cavity *Q* is switched high, which results in the generation of a lasing pulse.

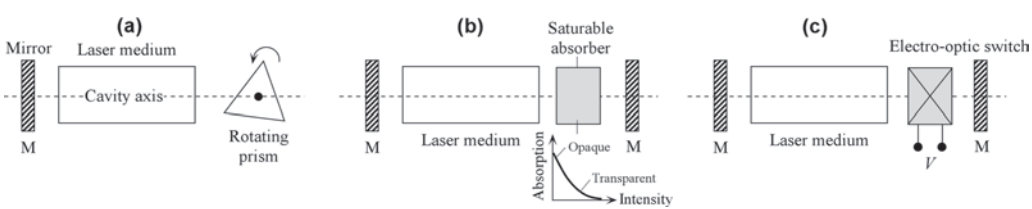


FIGURE 4.30 (a) *Q*-switching by using a rotating prism. (b) *Q*-switching by using a saturable absorber. (c) *Q*-switching by using an electro-optic (EO) switch. Normally a polarizer is also needed before or after the switch but this is part of the EO switch in this diagram.

B. Mode Locking

Mode locking is used to generate short and intense laser light pulses at a certain repetition rate that depends on the laser construction. A **mode-locked laser** is a laser that has been constructed to have one transverse mode and many (N) longitudinal modes, as in Figure 4.31 (a), that have the *same phase*. The N longitudinal modes then *reinforce* each other to generate an intense pulse of lasing emission at fixed time intervals as illustrated in Figure 4.31 (b). The repetition rate depends on the laser cavity length L and the speed of light in the lasing medium, c, assuming that the refractive index n is nearly 1 (*e.g.*, we have a gas laser). Normally, the longitudinal modes of a laser cavity would be "independent" with random relative phases. In such a case, the output intensity from the laser would simply be the sum of the individual mode intensities, *i.e.,* $I_1 + I_2 + \cdots + I_N$. If, on the other hand, the modes have been forced to have the same phase, then these mode oscillations would *reinforce* each other, *the optical fields would add*, in such a way that they generate an intense optical pulse at a certain repetition rate. In this mode-locked case, the intensity in a pulse is proportional to $(E_1 + E_2 + \cdots + E_N)^2$, and can be enormously larger than the case when the modes are not locked, as indicated in Figure 4.31 (b) where the horizontal dashed line, CW, represents the intensity in the unlocked laser. The repetition rate in a mode-locked laser is the reciprocal of the round-trip time of a light pulse in the optical cavity.

Suppose that $T = 2L/c$ is the round-trip time of a light pulse in the optical resonator shown in Figure 4.31 (c), where L is the cavity length and c is the velocity of light. The pulse repeats itself every $T = 2L/c$ seconds. The situation is analogous to taking a fixed number of sine waves with multiple frequencies, adjusting their phases so that one can obtain the maximum amplitude from their summation. The achievement of such an intense light pulse at every T seconds requires that the "modes are locked," that is, the relative phases of the modes have been correctly adjusted and fixed to yield the required maximum output intensity. When modes have been locked, there must be an optical pulse in the resonator that is traveling between the mirrors with exactly the required round-trip time of $T = 2L/c$. (If the refractive index n is not unity, we need to use c/n instead of c.)

The width Δt of an individual light pulse depends on the frequency width $\Delta\upsilon$ of the laser optical gain curve; normally $\Delta\upsilon$ is taken as the full width half maximum width. We know that $\Delta\upsilon$ also determines how many modes N are there in the output. The pulse width $\Delta t \approx 1/\Delta\upsilon \approx T/N$. In practice, it is not very difficult to obtain mode locking since the required output must correspond

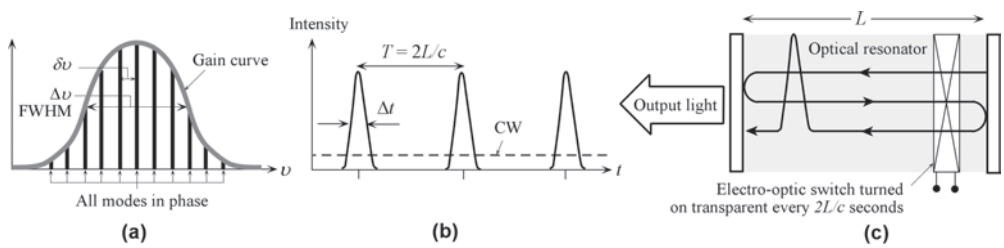


FIGURE 4.31 (a) A mode-locked laser has its N modes all in phase so that the modes add correctly to generate a short laser pulse every T seconds. $\Delta\upsilon$ is the full width at half maximum. (b) The output light intensity from a mode-locked laser is a periodic series of short intense optical pulses that are separated in time by $T = 2L/c$, the round-trip time for the pulse in the resonator. (c) A laser can be mode-locked by using an EO switch in the optical cavity that becomes transparent exactly at the right time, every T seconds. Each time the pulse in the resonator impinges on the left mirror, every $T = 2L/c$ seconds, a portion of it is transmitted, which constitutes the output from a mode-locked laser.

to an intense pulse train of repetition rate $1/T$. This output itself corresponds to a single optical pulse in the resonator bouncing back and forward between the mirrors with a round-trip time T. Each time this pulse impinges on a partially reflecting mirror, a portion of it is transmitted as an output pulse as in Figure 4.31 (c). Suppose that we insert an *electro-optic switch* that is switched to be made transparent at every T seconds (as we did for Q-switching). The switch is on to be transparent only when the pulse is there and only for the duration of the pulse. Then, the only possible situation is where a single optical pulse can bounce back and forward in the resonator and this corresponds to locking the modes. This is an example of **active mode locking**. The difference from the electro-optically Q-switched laser is that in mode locking, the EO switch must be turned on every T seconds for a very short time, so the timing in sequence is critical.

Another possibility is **passive mode locking** where a saturable absorber is used in the optical cavity (again, similar to Q-switching). The optical cavity is lossy for low light intensities. Such an absorber would only allow a high intensity light pulse to exist in the cavity since it is only transparent at high intensities. The latter would correspond to various modes having the right phases to yield an intense pulse, that is, mode locking. The difference from the Q-switching case is that the absorber in a mode-locked laser must be able to respond faster than the time T.

The output pulses from a mode-locked laser are spatially separated by $2L$ because the pulses come out at every T seconds from the output mirror and Tc is $2L$. A mode-locked He-Ne laser typically has a pulse width of roughly 600 ps whereas it is 150 fs for a mode-locked Nd^{3+}: glass laser. It might be thought that we have generated a more powerful output laser beam, a stream of high intensity pulses, than the CW (unlocked) case, but this is not true. The peak power in an individual mode-locked pulse, roughly the energy in the pulse divided by the pulse width Δt, can be extremely high, often in megawatts, but the time-averaged power (over many pulses) is the same as that in the CW unlocked laser.

4.9 PRINCIPLE OF THE LASER DIODE[28]

Consider a degenerately doped direct bandgap semiconductor *pn* junction whose band diagram is shown in Figure 4.32 (a). By degenerate doping we mean that the Fermi level E_{Fp} in the p-side is in the valence band (VB) and E_{Fn} in the n-side is in the conduction band (CB). All energy levels up to the Fermi level can be taken to be occupied by electrons as in Figure 4.32 (a). In the absence of an applied voltage, the Fermi level is continuous across the diode, $E_{Fp} = E_{Fn}$. The depletion region or the space charge layer (SCL) in such a *pn* junction is very narrow. There is a built-in voltage V_o that gives rise to a potential energy barrier eV_o that prevents electrons in the CB of n^+-side diffusing into the CB of the p^+-side.[29] There is a similar barrier stopping the hole diffusion from p^+-side to n^+-side.

Recall that when a voltage is applied to a *pn* junction device, the change in the Fermi level from end-to-end is the electrical work done by the applied voltage, that is, $\Delta E_F = eV$. Suppose

[28] The first semiconductor lasers used GaAs *pn* junctions, as reported by the American researchers Robert N. Hall *et al.* (General Electric Research, Schenectady), Marshall I. Nathan *et al.* (IBM, Thomas J. Watson Research Center), and Robert H. Rediker *et al.* (MIT), in 1962. Nick Holonyak (General Electric at Syracuse), also in 1962, reported a laser diode based on the compound GaAsP, which emitted in the visible (red).

[29] The potential energy barrier eV_o in a nondegerate *pn* junction is normally taken as the energy required for an electron to move from E_c on the n-side to E_c on the p-side since electrons are near the bottom of the CB at E_c. In the present case, electrons occupy states from E_c up to E_{Fn}. The exact analysis of the problem is therefore more complicated; however $E_{Fn} - E_c$ is small compared to eV_o.

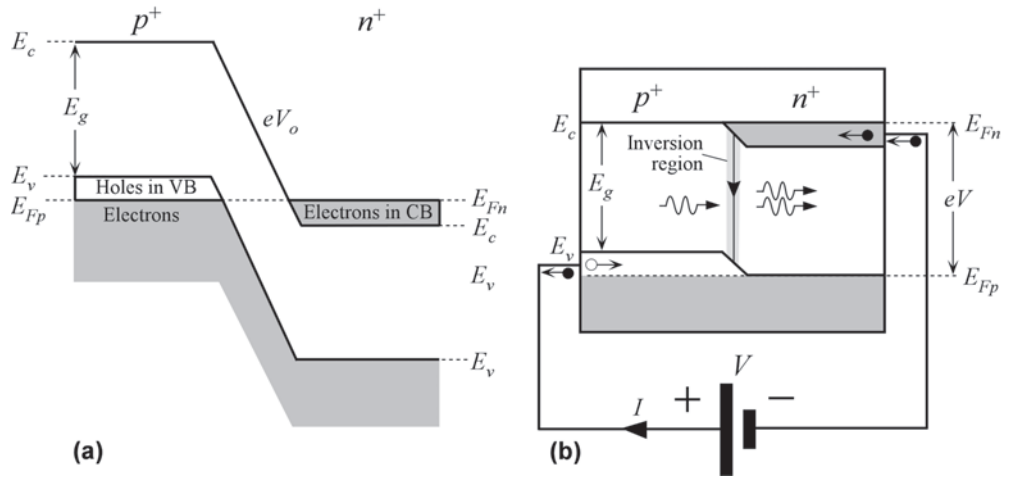


FIGURE 4.32 (a) The energy band diagram of a degenerately doped *pn* with no bias. eV_o is the potential energy barrier against electron diffusion from the n^+-side to the p^+-side. Note that $(E_{Fn} - E_c)$ and $(E_v - E_{Fp})$ are small compared with eV_o. (The diagram is exaggerated.) (b) Band diagram with a sufficiently large forward bias to cause population inversion and hence stimulated emission. (As before, filled black circles are electrons and empty circles are holes. Only electrons flow in the external circuit.)

that this degenerately doped *pn* junction is forward biased by a voltage V greater than the band-gap voltage; $eV > E_g$ as shown in Figure 4.32 (b). The separation between E_{Fn} and E_{Fp} is now the applied potential energy or eV. The applied voltage diminishes the built-in potential barrier to almost zero, which means that electrons flow into the SCL and flow over to the p^+-side to constitute the diode current. There is a similar reduction in the potential barrier for holes from p^+- to n^+-side. The final result is that electrons from n^+-side and holes from p^+-side flow into the SCL, and this SCL region is no longer depleted, as apparent in Figure 4.32 (b). If we draw the energy band diagram with $E_{Fn} - E_{Fp} = eV > E_g$ this conclusion is apparent. In this region, there are *more* electrons in the conduction band at energies near E_c than electrons in the valence band near E_v as illustrated by density of states diagram for the junction region in Figure 4.33 (a). In other words, there is a **population inversion** between energies near E_c and those near E_v around the junction.

This population inversion region is a layer along the junction and is called the **inversion layer** or the **active region**. An incoming photon with an energy of $(E_c - E_v)$ cannot excite an

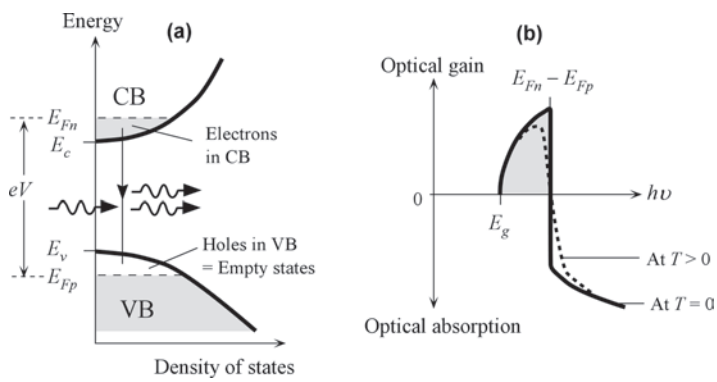


FIGURE 4.33 (a) The density of states and energy distribution of electrons and holes in the conduction and valence bands, respectively, in the SCL under forward bias such that $E_{Fn} - E_{Fp} > E_g$. Holes in the VB are empty states. (b) Gain vs. photon energy ($h\upsilon$). The effect of temperature is also shown.

electron from E_v to E_c as there are almost none near E_v. It can, however, *stimulate* an electron to fall down from E_c to E_v as shown in Figure 4.32 (b). Put differently, the incoming photon stimulates direct recombination. The region where there is population inversion and hence more stimulated emission than absorption, or the active region, has an **optical gain** because an incoming photon is more likely to cause stimulated emission than being absorbed. The optical gain depends on the photon energy (and hence on the wavelength) as apparent by the energy distributions of electrons and holes in the conduction and valence bands in the active layer in Figure 4.33 (a). At low temperatures ($T \approx 0$ K), the states between E_c and E_{Fn} are filled with electrons and those between E_{Fp} and E_v are empty. Photons with energy greater than E_g but less than $E_{Fn} - E_{Fp}$ cause stimulated emissions, whereas those photons with energies greater than $E_{Fn} - E_{Fp}$ become absorbed. Figure 4.33 (b) shows the expected dependence of optical gain and absorption on the photon energy at low temperatures ($T \approx 0$ K). As the temperature increases, the Fermi–Dirac function spreads the energy distribution of electrons in the CB to above E_{Fn} and holes below E_{Fp} in the VB. Put differently, electrons from below E_{Fn} are spread to energies above E_{Fn}. The result is a reduction and also some smearing of the optical gain curve as indicated in Figure 4.33 (b). The optical gain depends on $E_{Fn} - E_{Fp}$, which depends on the applied voltage, and hence on the diode current.

It is apparent that population inversion between energies near E_c and those near E_v is achieved by the injection of carriers across the junction under a sufficiently large forward bias. The pumping mechanism is therefore the forward diode current and the pumping energy is supplied by the external battery. This type of pumping is called **injection pumping**.

In addition to population inversion we also need to have an *optical cavity* to implement a laser oscillator, that is, to build up the intensity of stimulated emissions by means of an optical resonator. This would provide a continuous coherent radiation as output from the device. Figure 4.34 shows schematically the structure of a **homojunction laser diode**. The *pn* junction uses the same direct bandgap semiconductor material throughout, for example GaAs, and hence has the name homojunction. The ends of the crystal are cleaved to be flat and optically polished to provide reflection and hence form an optical cavity. Photons that are reflected from the cleaved surfaces stimulate more photons of the same frequency and so on. This process builds up the intensity of the radiation in the cavity. The wavelength of the radiation that can build up in the cavity is determined by the length L of the cavity because

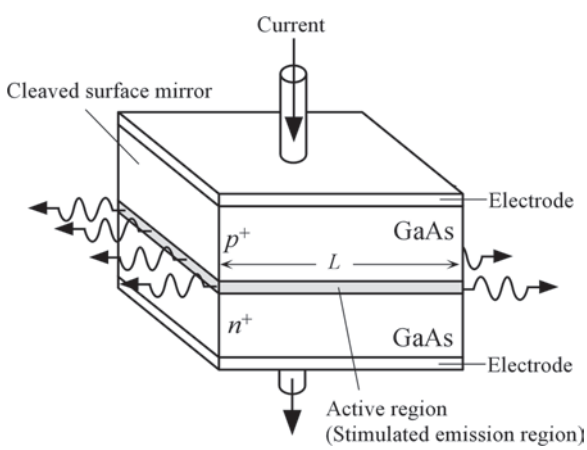
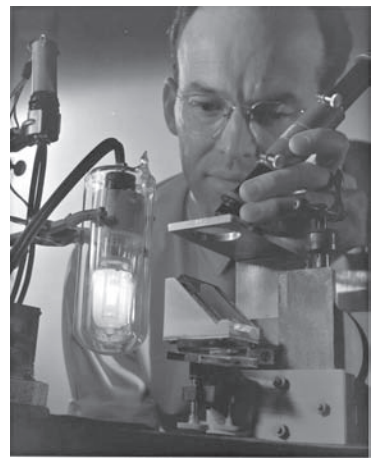


FIGURE 4.34 A schematic illustration of a GaAs homojunction laser diode. The cleaved surfaces act as reflecting mirrors.



Robert Hall and his colleagues, while working at General Electric's Research and Development Center in Schenectady, New York, were among the first groups of researchers to report a working semiconductor laser diode in 1962. He obtained a U.S. patent in 1967, entitled "Semiconductor junction laser diode" for his invention. When Robert Hall retired from GE in 1987, he had been awarded more than 40 patents. *(*R. N. Hall et al., *Phys. Rev. Letts., 9,* 366, 1962; *Courtesy of GE.)*

only multiples of the half-wavelengths can exist in such an optical cavity as explained in Section 4.6D, *i.e.,*

Modes in an optical cavity

$$m\frac{\lambda}{2n} = L \tag{4.9.1}$$

where, as before, m is an integer, called a **mode number**, n is the refractive index of the semiconductor, and λ is the free-space wavelength. Each radiation pattern satisfying the above relationship is essentially a **resonant frequency** of the cavity, that is, a **mode** of the cavity. The separation between possible modes of the cavity (or separation between allowed wavelengths) $\Delta\lambda_m$ can be readily found from Eq. (4.9.1) as in the case of the He-Ne gas laser previously.

The dependence of the optical gain of the medium on the wavelength of radiation can be deduced from the energy distribution of the electrons in the CB and holes in the VB around the junction as in Figures 4.33 (a) and (b). The exact output spectrum from the laser diode depends both on the nature of the optical cavity and the optical gain vs. wavelength characteristics. Lasing radiation is only obtained when the optical gain in the medium can overcome the photon losses from the cavity, which requires the diode current I to exceed a certain threshold value I_{th}. (This is similar to the condition we had for gas lasers where pumping must exceed some threshold pumping required for the threshold gain.) Below I_{th}, the light from the device is due to spontaneous emission and not stimulated emission. The light output is then composed of incoherent photons that are emitted randomly and the device behaves like an LED. For diode currents above I_{th}, the device emits coherent lasing emission.

We can identify two critical diode currents. First is the diode current that provides just sufficient injection to lead to stimulated emissions just balancing absorption. This is called the **transparency current** I_T inasmuch as there is then no net photon absorption; the medium is perfectly *transparent.* Above I_T there is optical gain in the medium though the optical output is not yet a continuous wave coherent radiation. Lasing oscillations occur only when the optical gain in the medium can overcome the photon losses from the cavity, that is, when the optical gain g reaches the threshold gain g_{th}. This occurs at the **threshold current** I_{th}. Those cavity resonant frequencies that experience the threshold optical gain can resonate within the cavity. Some of this cavity radiation is transmitted out from the cleaved ends as these are not perfectly reflecting (typically about

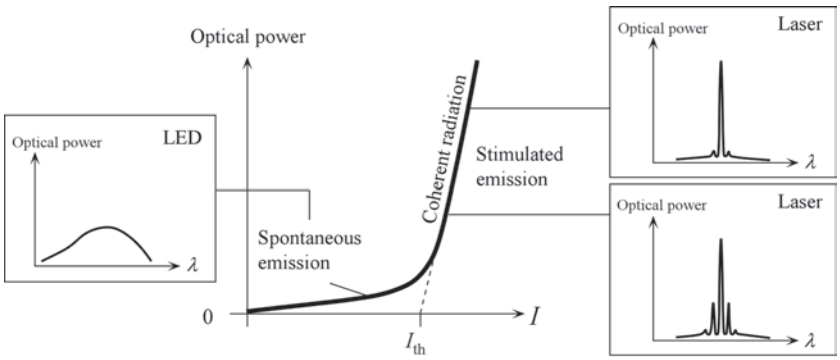


FIGURE 4.35 Typical output optical power vs. diode current (I) characteristics and the corresponding output spectrum of a laser diode. I_{th} is the threshold current and corresponds to the extension of the coherent radiation output characteristic onto the I-axis.

32% reflecting without any antireflection coating). Figure 4.35 shows the output light intensity as a function of diode current. Below the threshold current I_{th}, the light output is incoherent radiation due to the spontaneous recombination of injected electrons and holes. Above I_{th}, the light output becomes coherent radiation consisting of cavity wavelengths (or modes); and the output intensity increases steeply with the current. The number of modes in the output spectrum and their relative strengths depend on the diode current as illustrated in Figure 4.35. Notice that as the current increases, one of the intense modes becomes prominent at the expense of less intense modes. The prominent mode has so much intensity that it ends up using most of the injected electrons; there are insufficient electrons left for other modes to satisfy the threshold gain condition at those frequencies.

The main problem with the homojunction laser diode is that the threshold current density J_{th} is too high for practical uses. For example, the threshold current density is of the order of $\sim 500\,\text{A mm}^{-2}$ for GaAs *pn* junctions at room temperature, which means that the GaAs homojunction laser can be operated continuously only at very low temperatures. However, J_{th} can be reduced by orders of magnitude by using **heterostructure** semiconductor laser diodes.

4.10 HETEROSTRUCTURE LASER DIODES

The reduction of the threshold current I_{th} to a practical value requires improving the rate of stimulated emission and also improving the efficiency of the optical cavity. First, we can confine the injected electrons and holes to a narrow region around the junction. This narrowing of the active region means that less current is needed to establish the necessary concentration of carriers for population inversion. Second, we can build a dielectric waveguide around the optical gain region to increase the photon concentration and hence the probability of stimulated emission. This way we can reduce the loss of photons traveling off the cavity axis. We therefore need both **carrier confinement** and **photon** or **optical confinement.** Both of these requirements are readily achieved in modern laser diodes by the use of heterostructured devices as in the case of high-intensity double heterostructure LEDs. However, in the case of laser diodes, there is an additional requirement for maintaining a good optical cavity that will increase stimulated emission over spontaneous emission.

Figure 4.36 (a) shows a **double heterostructure** (DH) device based on two junctions between different semiconductor materials with different bandgaps. In this case, the semiconductors are $Al_{1-x}Ga_xAs$ (or simply AlGaAs) with $E_g \approx 2$ eV and GaAs with $E_g \approx 1.4$ eV. The *p*-GaAs region is a thin layer, typically about 0.1 μm, and constitutes the **active layer** in which



Izuo Hayashi (left) and Morton Panish (1971) at Bell Labs were able to design the first semiconductor laser that operated continuously at room temperature. The need for semiconductor heterostructures for efficient laser diode operation was put forward by Herbert Kroemer in the United States and Zhores Alferov in Russia in 1963. (Notice the similarity of the energy band diagram on the chalkboard with that in Figure 4.36.) *(Reprinted with permission of Alcatel–Lucent USA Inc.)*

lasing recombination takes place. Both p-GaAs and p-AlGaAs are heavily p-type doped and are degenerate with E_F in the valence band. When a sufficiently large forward bias is applied, E_c of n-AlGaAs moves above E_c of p-GaAs, which leads to a large injection of electrons from the CB of n-AlGaAs into the CB of p-GaAs as shown in Figure 4.36 (b). These electrons, however, are *confined* to the CB of p-GaAs since there is a potential barrier ΔE_c between p-GaAs and p-AlGaAs due to the change in the bandgap (there is also a small change in E_v but we ignore this). Inasmuch as p-GaAs is a thin layer, the concentration of injected electrons in the p-GaAs layer can be increased quickly even with moderate increases in forward current. This effectively reduces the threshold current for population inversion or optical gain. Thus, even moderate forward currents can inject sufficient number of electrons into the CB of p-GaAs to establish the necessary electron concentration for population inversion in this layer.

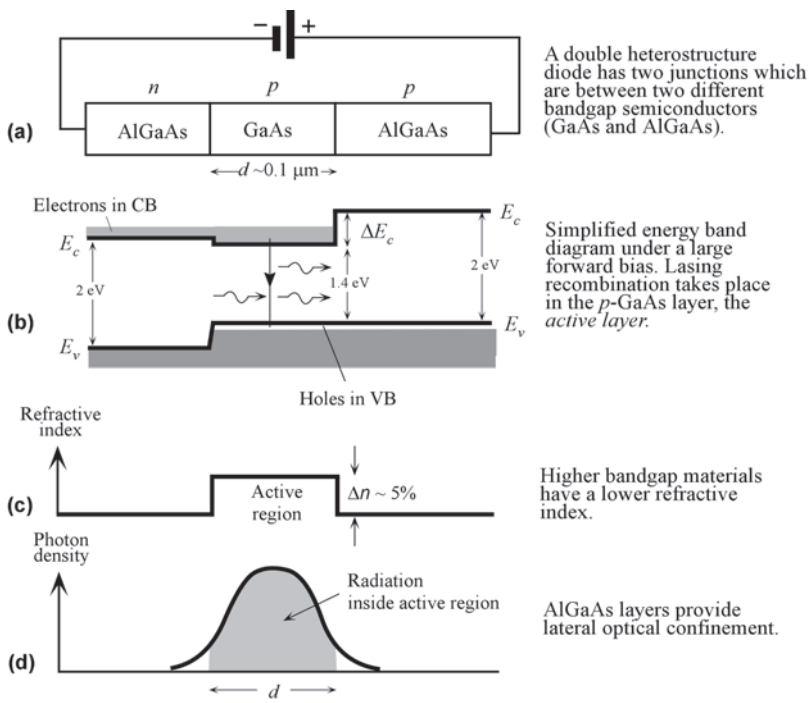


FIGURE 4.36 The basic principle of operation of a double heterostructure laser.

A wider bandgap semiconductor generally has a lower refractive index. AlGaAs has a lower refractive index than that of GaAs. The change in the refractive index defines an optical dielectric waveguide, as illustrated in Figure 4.36 (c), that confines the photons to the active region of the optical cavity and thereby increases the photon concentration. The photon concentration across the device is shown in Figure 4.36 (d). This increase in the photon concentration increases the rate of stimulated emissions. Thus, both carrier and optical confinement lead to a reduction in the threshold current density. Without double heterostructure devices we would not have practical solid state lasers that can be operated continuously at room temperature.

A typical structure of a **double heterostructure laser diode** is similar to a double heterostructure LED and is shown schematically in Figure 4.37. The doped layers are grown epitaxially on a crystalline substrate which in this case is *n*-GaAs. The double heterostructure described above consists of the first layer on the substrate, *n*-AlGaAs, the active *p*-GaAs layer, and the *p*-AlGaAs layer. There is an additional *p*-GaAs layer, called **contacting layer**, next to *p*-AlGaAs. It can be seen that the electrodes are attached to the GaAs semiconductor rather than AlGaAs. This choice allows for better contacting, that is, smaller contact resistance. The *p*- and *n*-AlGaAs layers provide carrier and optical confinement in the vertical direction by forming heterojunctions with *p*-GaAs. The active layer is *p*-GaAs, which means that the lasing emission will be in the range 870–900 nm depending on the doping level. This layer can also be made to be $Al_yGa_{1-y}As$ but of different composition than the confining $Al_xGa_{1-x}As$ layers, and still preserve heterojunction properties. By modifying the composition of the active layer, we can control the wavelength of the lasing emission over 650–900 nm. The advantage of AlGaAs/GaAs heterojunction is that there is only a small lattice mismatch between the two crystal structures and hence negligible strain induced interfacial defects (*e.g.*, dislocations) in the device. Such defects invariably act as nonradiative recombination centers and hence reduce the rate of radiative transitions.

An important feature of this laser diode is the **stripe geometry**, or stripe contact on *p*-GaAs. The current density *J* from the stripe contact is not uniform laterally. *J* is greatest

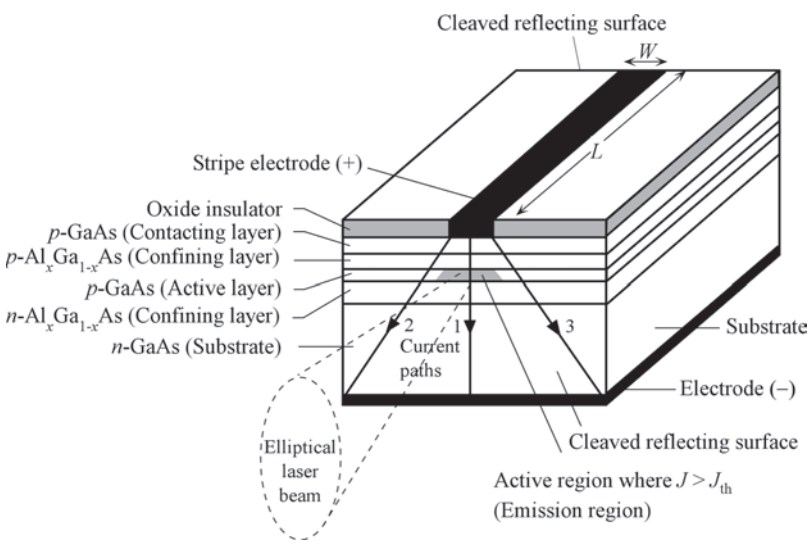


FIGURE 4.37 Schematic illustration of the structure of a double heterojunction stripe contact laser diode.

The peak in the frequency response at f_r is typical of a system that has two energy storage elements, such as an inductor L and a capacitor C that are coupled. We can view the LD as made up of two subsystems. One subsystem is the electrical subsystem that has the current and the injected electrons, and the other, the optical subsystem, that has the optical cavity and the coherent radiation. Near f_r, the energy associated with the injected electrons in the active region in the electrical subsystem is transferred back and forth to and from the radiation (photons) within the cavity in the optical subsystem. This energy transfer is mediated by stimulated emissions because both photons and electrons are involved in stimulated transitions. The electron and photon concentrations are coupled through the stimulated emission process. The phase difference ϕ between the output and the input signals arises from the delay in getting the coherent photon output keeping up with the electron injection. There is a delay associated with the recombination process, which is represented by τ, and then there is another delay associated with getting the photons out from the cavity τ_{ph}. The radiation output lags the input current.

Consider a LD that is excited by a step current pulse that changes from I_1 ($<I_{th}$) to I_2 ($>I_{th}$) as shown in Figure 4.56 (a). Assume initially I_1 is very small. The lasing radiation output begins only after a **delay time** t_d, and reaches a steady state value after a few damped oscillations as illustrated in Figure 4.56 (b). The delay time t_d is the time it takes to inject sufficient electrons into the active region to achieve population inversion, and sufficient gain to start the lasing oscillations to build up. Suppose that initially the bias on the LD is I_1, less than I_{th}, as in Figure 4.56 (a). Thus, the diode current changes from I_1 to I_2. The delay time t_d before any coherent radiation output appears is approximately given by

$$t_d \approx \tau \ln\left[\frac{I_2}{I_2 - (I_{th} - I_1)}\right] \qquad \textbf{(4.18.2)}$$

Diode laser output delay time

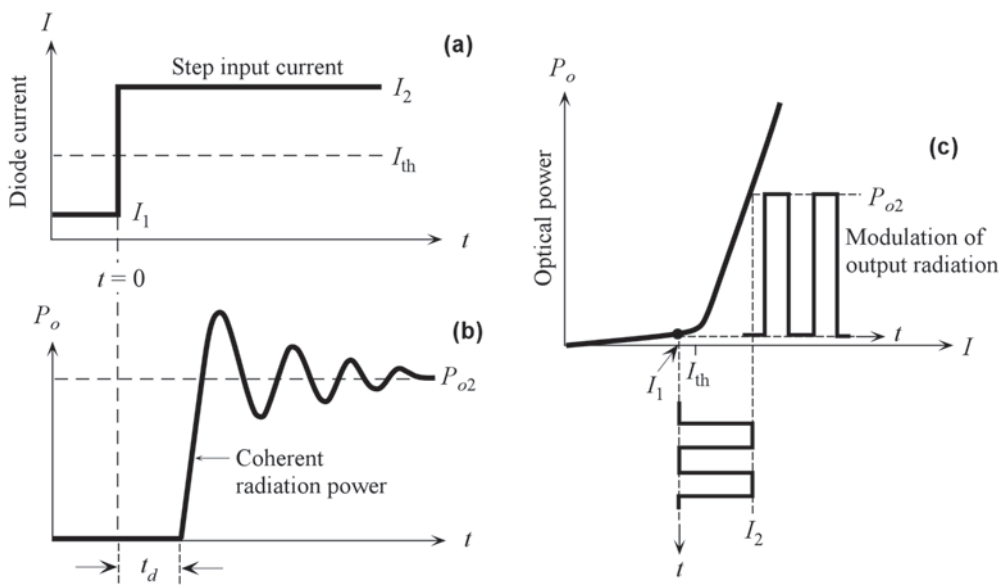


FIGURE 4.56 (a) The Diode current is applied suddenly at $t = 0$ as a step from I_1 (below I_{th}) to I_2 (above I_{th}). (b) The transient response of the output power, P_o, as a function of time. There is a delay t_d before any output appears. P_{o2} is the steady state output, after the relaxation oscillations have died out, after a few cycles. (c) The laser diode is biased at I_1 near I_{th} for faster switching. The input current is a pulse of magnitude $(I_2 - I_1)$, which generates an output P_{o2}. (Ideal response in which time delays have been neglected, as would be the case under sufficiently low repetition rates.)

where τ is the overall effective recombination time in the recombination region around the threshold operating conditions and $(I_{th} - I_1)$ is the current needed to reach I_{th}. If the LD is not biased, the delay time will be longer. We can reduce the delay time by biasing the LD at I_1 just below I_{th}, as shown in Figure 4.56 (c), so that $(I_{th} - I_1)$ is negligible, then t_d is as short as possible and roughly the same as τ, the recombination time. The damped relaxation oscillations in the output in Figure 4.56 (b) have the frequency f_r, which should not be surprising, given the above explanation for f_r in terms of two energy storage subsystems in the LD. Usually, the output reaches a steady state P_{o2} after a few oscillations.

Generally, under direct modulation, LD bandwidths are wider than conventional LEDs, and LDs are the obvious choice in high-speed optical communications, given also their much narrower spectral width. There is also a disadvantage to directly modulating LDs by the diode current. The modulation processes modifies the refractive index of the gain medium, and hence causes the radiation frequency to shift, called **frequency chirping**, during the modulation period, over the time of a digital bit. The reason is that the refractive index n, as mentioned above, depends on the injected electron concentration, which is controlled by the current. The frequency chirp can be eliminated by operating the LD as a CW laser, and using a high-speed external modulator. Some electro-optic external modulators, such as Mach–Zhender modulators, as discussed in Chapter 6, have very fast switch times and can easily operate at 40 Gb s^{-1} or above.

4.19 HOLOGRAPHY

Holography is a technique of reproducing three-dimensional optical images of an object by using a highly coherent radiation from a laser source. Although holography was invented by Denis Gabor in 1948, prior to the availability of highly coherent laser beams, its practical use and popularity increased soon after the commercial development of lasers.

The principle of holography is illustrated in Figures 4.57 (a) and (b). The object, a cat, is illuminated by a highly coherent beam as normally would be available from a laser[42] as in Figure 4.57 (a). The laser beam has both *spatial* and *temporal coherence.* Part of the coherent wave is reflected from a mirror to form a **reference beam** E_{ref} and travels toward a fine-grained photographic plate. The waves that are reflected from the cat, E_{cat}, will have both amplitude and phase variations that represent the cat's surface (topology). If we were looking at the cat, our eyes would register the wavefront of the reflected waves E_{cat}. Moving our head around, we would capture different portions of the reflected wave and we would see the cat as a three-dimensional object.

The reflected waves from the cat (E_{cat}) is made to interfere with a reference wave (E_{ref}) at the photographic plate and give rise to a complicated *interference pattern* that depends on the magnitude and phase variation in E_{cat}. The recorded interference pattern in the photographic film (after processing) is called a **hologram**. It contains all the information necessary to reconstruct the wavefront E_{cat} reflected from the cat and hence produce a three-dimensional image.

To obtain the three-dimensional image, we have to illuminate the hologram with the reference beam E_{ref} alone as in Figure 4.57 (b). Most of the beam goes right through but some of it becomes *diffracted* by the interference pattern in the hologram. One diffracted beam is an exact replica of the original wavefront E_{cat} from the cat. The observer sees this wavefront as if the waves were reflected from the original cat and registers a three-dimensional image of the cat. This is the **virtual image**. We know that first-order diffraction from a grating has to satisfy the

[42] A short pulse of laser light, such as from a ruby laser, will not harm the cat; this is how holograms of people are taken. The first successful three-dimensional laser holography was carried out by Emmett Leith and Juris Upatneiks (University of Michigan) in 1960. In 1979 President Jimmy Carter awarded Emmett Leith with the National Medal of Science for his contributions to optics.

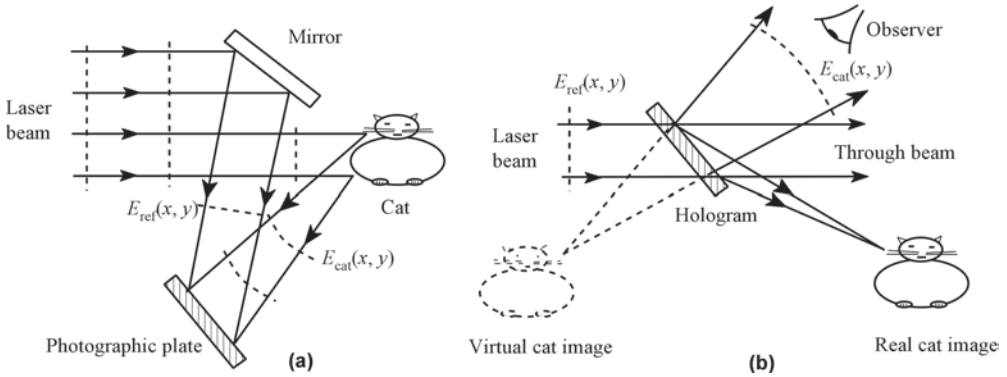


FIGURE 4.57 A highly simplified illustration of holography. (a) A laser beam is made to interfere with the diffracted beam from the subject to produce a hologram. (b) Shining the laser beam through the hologram generates a real and a virtual image.

Bragg condition, $d\sin\theta = m\lambda\,(m = \pm 1)$ where λ is the wavelength and d is the periodicity of the grating, which we took as the separation of slits in a ruled grating in Chapter 1. We can qualitatively think of a diffracted beam from one locality in the hologram as being determined by the local separation d between interference fringes in this region. Since d changes in the hologram, depending on the interference pattern produced by E_{cat}, the whole diffracted beam depends on E_{cat} and the diffracted beam wavefront is an exact scaled replica of E_{cat}. Just as normally there would be another diffracted beam on the other side of the zero-order (through) beam, there is a second image, called the **real image**, as in Figure 4.57 (b), which is of lower quality. (It may help to imagine what happens if the object consists of black and white stripes. We would then obtain periodic dark and bright interference fringes in the hologram. This periodic variation is just like a diffraction grating; the exact analysis is more complicated.)

Holography can be explained by the following highly simplified analysis. Suppose that the photographic plate is in the xy plane in Figure 4.57. Assume that the reference wave can be represented by

$$E_{ref}(x, y) = U_r(x, y)e^{j\omega t} \tag{4.19.1}$$

where $U_r(x, y)$ is its amplitude, generally a complex number that includes magnitude and phase information.

The reflected wave from the cat will have a complex amplitude that contains the magnitude and phase information of the cat's surface, that is,

$$E_{cat}(x, y) = U(x, y)e^{j\omega t} \tag{4.19.2}$$

where $U(x, y)$ is a complex number. Normally we would look at the cat and construct the image from the wavefront in Eq. (4.19.2). We thus have to reconstruct the wavefront in Eq. (4.19.2), *i.e.,* obtain $U(x, y)$.

When E_{ref} and E_{cat} interfere, the "brightness" of the photographic image depends on the intensity I and hence on

$$I(x, y) = |E_{ref} + E_{cat}|^2 = |U_r + U|^2 = (U_r + U)(U_r^* + U^*)$$

where we have used the usual definition that the square of the magnitude of a complex number is the product with its complex conjugate (indicated by an asterisk). Thus,

$$I(x, y) = UU^* + U_rU_r^* + U_r^*U + U_rU^* \tag{4.19.3}$$

This is the pattern on the photographic plate. The first and second terms are the intensities of the reflected and reference waves. It is the third and fourth terms that contain the information on the magnitude and phase of $U(x, y)$.

We now illuminate the hologram, $I(x, y)$, with the reference beam, which has an amplitude U_r as in Eq. (4.19.1), so that the transmitted light wave has a complex magnitude $U_t(x, y)$ proportional to $U_r I(x, y)$

$$U_t \propto U_r I(x, y) = U_r(UU^* + U_r U_r^* + U_r^* U + U_r U^*)$$

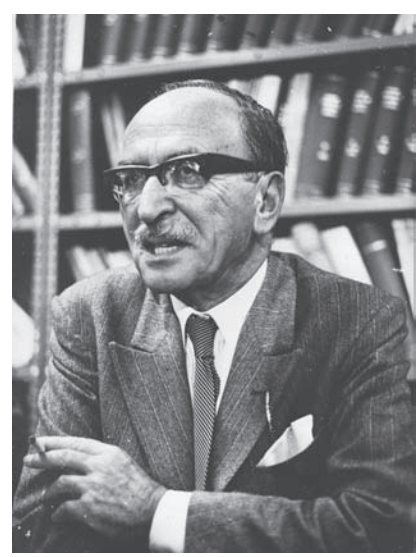
i.e.,

$$U_t \propto U_r(UU^* + U_r U_r^*) + (U_r U_r^*)U + U_r^2 U^* \tag{4.19.4}$$

or

$$U_t \propto a + bU(x, y) + cU^*(x, y)$$

where a, b, and c are appropriate constants. The first term $a = U_r(UU^* + U_r U_r^*)$ is the through beam. The second and third terms represent the diffracted beams.[43] Since $b = (U_r U_r^*)$ is a constant (intensity of the reference beam), the second term is a scaled version of $U(x, y)$ and represents the original wavefront amplitude from the cat, *i.e.,* includes the magnitude and phase information. We have effectively reconstructed the original wavefront and the observer will see a three-dimensional view of the cat within the angle of original recording; $bU(x, y)$ is the **virtual image**. The third term $cU^*(x, y)$ is the **real image** and notice that its amplitude is the complex conjugate of $U(x, y)$; it is called the **conjugate image**. It is apparent that holography is a method of **wavefront reconstruction**. It should be mentioned that the observer will always see a positive image whether a positive or a negative image of the hologram is used. A negative hologram simply produces a 180° phase shift in the field of the transmitted wave and the eye cannot detect this phase shift.



Dennis Gabor (1900–1979), inventor of holography, was a Hungarian-born British physicist who published his holography invention in *Nature* in 1948 while he was at Thomson-Houston Co. Ltd, at a time when coherent light from lasers was not yet available. He was subsequently a professor of applied electron physics at Imperial College, University of London. He received the Nobel Prize in Physics in 1971 for his invention of holography. His Nobel Lecture given in 1971 provides an excellent insight into the development of holography, and is accessible from the Nobel website. *(© Linh Hassel/AGE Fotostock.)*

[43] This is a simple factual statement as the mathematical treatment above has not proved this point.

Questions and Problems

4.1 Pumping of three-level laser systems The 3-level Ruby laser system is shown in Figure 4.2 (a). If transitions from E_3 to E_2 are very fast, and the spontaneous decay time from E_2 to E_1 is τ_{sp}, obtain an expression for the minimum pumping power needed (the power that must be absorbed by the laser medium) for population inversion ($N_2 > N_1$).

Consider a 60 mm long ruby crystal with 6 mm diameter, doped with $2 \times 10^{20}\,Cr^{3+}$ ions cm^{-3}. The spontaneous decay time is 3 ms. Using the above-obtained expression, find the minimum power that must be absorbed if the excitation is to the band of energies in the blue around (~3.1 eV) in Figure 4.3 (a).

4.2 Neodymium-doped glass laser Consider a phosphate glass rod, with a refractive index $n = 1.50$, that has been doped with $3 \times 10^{20} Nd^{3+}$ ions cm^{-3}. The glass rod is 10 cm long and 1 cm in diameter. We can use Figure 4.4 to represent the operation of the Nd^{3+}:glass laser. The lasing emission E_2 to E_1 is at 1054 nm. The pump wavelength is 808 nm, and takes the Nd^{3+} from E_0 to E_3. E_1 is roughly 0.26 eV above E_0. The linewidth of the output spectrum $\Delta\lambda$ is about 28 nm. E_2 to E_1 emission cross-section is 3×10^{-20} cm^2 and the spontaneous decay lifetime is 300 μs. (a) Calculate the maximum possible gain coefficient in the medium. (b) What percentage of the input energy is lost as heat? (c) What is the spectral spread (linewidth) in terms of photon energy?

4.3 Erbium doped fiber amplifier A glass fiber has been doped with Er^{3+} ions with doping concentration $8 \times 10^{18}\,Er^{3+}$ ions cm^{-3}. Emission and absorption cross-sections of this EDFA at a wavelength of 1550 nm are found to be 3×10^{-21} cm^2 and 2×10^{-21} cm^2, respectively. The Er^{3+} doped fiber is pumped from a 1480 nm laser diode. Obtain the maximum gain coefficient in terms of dB m^{-1} of this medium at 1550 nm. You may assume that the maximum gain coefficient is obtained when there is full population inversion ($N_1 = 0$).

4.4 Erbium-doped fiber amplifier Consider an EDFA that is pumped at 1480 nm. Let N_1 and N_2 be the concentrations of Er^{3+} at E_2 and E_1 manifolds in Figure 4.7 (b). $N_1 + N_2 = N =$ Concentration of Er^{3+} ions, taken as 10^{19} cm^{-3}. When transparency is achieved (no net absorption nor gain), $g = 0$, and $N_1 = N_{1T}$ and $N_2 = N_{2T}$. Show that

$$N_{2T} = N\sigma_{ab}(\upsilon)/\left[\sigma_{ab}(\upsilon) + \sigma_{em}(\upsilon)\right]$$

where $\sigma_{ab}(\upsilon)$ and $\sigma_{em}(\upsilon)$ are the absorption and emission cross-sections that depend on the frequency (wavelength) of interest. The 1480-nm pump generates N_{2T} of Er^{3+} in the manifold 2 in Figure 4.7 (b). The spontaneous decay time from E_2 to E_1 is τ_{sp}. The pump must accumulate the Er^{3+} ions at E_2 before they can decay. Show that the **minimum pump power needed for transparency** is

$$P_{pump} \approx ALh\upsilon_p N\sigma_{ab}/\tau_{sp}(\sigma_{ab} + \sigma_{em})$$

where υ_p is the pump frequency, A the pump core area, and L is the fiber length. Calculate the minimum pump power needed for an EDFA if $N = 10^{19}\,cm^{-3}$, $\tau_{sp} \approx 10$ ms, core diameter is 4 μm, the fiber length is 7 m, and the transparency is at 1550 nm where emission and absorption cross-sections are $4 \times 10^{-21}\,cm^2$ and $3 \times 10^{-21}\,cm^2$, respectively. What would be the power needed for pumping *all* Er^{3+} ions to E_2?

4.5 Erbium-doped fiber amplifier Consider an EDFA with an Er doping concentration of 2×10^{19} cm^{-3} and pumped at 1480 nm. The fiber length is 3 m. Find the small signal gain G at two wavelengths at 1550 and 1570 nm corresponding to 100% population inversion.

4.6 Erbium-doped fiber amplifier Consider a 5 m long EDFA with Er^{3+} doping concentration of $1 \times 10^{19}\,cm^{-3}$, core diameter of 8 μm. The spontaneous decay time τ_{sp} from E_2 to E_1 is 10 ms. The fiber is pumped at 980 nm from a laser diode. The pump power coupled into the EDFA fiber is 40 mW. What is the fiber length that will absorb the pump radiation when the confinement factor is 80%? Find the small signal gain at 1550 nm corresponding to 80% population inversion.

4.7 Erbium-doped fiber amplifier What is the maximum input signal power beyond which saturation effects take place for an EDFA having a small signal gain (G) of 30 dB, pumped at 80 mW at 980 nm?

4.8 He-Ne Laser The He-Ne laser system energy levels can be quite complicated, as shown in Figure 4.58. There are a number of lasing emissions in the laser output in the red (632.8 nm), green (543.5 nm), orange (612 nm), yellow (594.1 nm), and in the infrared regions at 1.52 μm and 3.39 μm, which give the He-Ne laser its versatility. The pumping mechanism for all these lasers operations is the same, energy transfer from excited He atoms to Ne atoms by atomic collisions in the gas discharge tube. There are two excited states for the He

atom in the configuration $1s^1 2s^2$. The configuration with electron spins parallel has a lower energy than that with opposite electron spins as indicated in Figure 4.58. These two He states can excite Ne atoms to either the $2p^5 4s^1$ or $2p^5 5s^1$. There is then a population inversion between these levels and the $2p^5 3p^1$ and between $2p^5 5s^1$ and $2p^5 4p^1$, which leads to the above lasing transitions. Generally, we do not have a single discrete level for the energy of a many-electron atom of given n,ℓ configuration. For example, the atomic configuration $2p^5 3p^1$ has ten closely spaced energy levels resulting from various different values of m_l and m_s that can be assigned to the sixth excited electron ($3p^1$) and the remaining electrons ($2p^5$) within quantum mechanical rules. Not all transitions to these energy levels are allowed as photon emission requires quantum number selection rules to be obeyed.

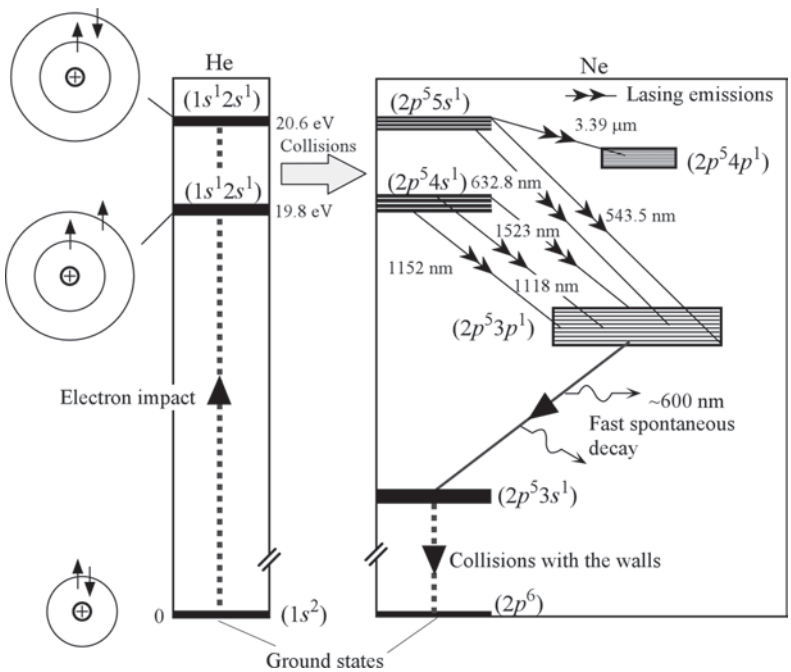


FIGURE 4.58 Various lasing transitions in the He-Ne laser.

(a) Table 4.8 shows some typical commercial He-Ne laser characteristics (within 30–50%) for various wavelengths. Calculate the overall efficiencies of these lasers.

TABLE 4.8 Typical commercial He-Ne laser characteristic

Wavelength (nm)	543.5	594.1	612	632.8	1523
	Green	Yellow	Orange	Red	Infrared
Optical output power (mW)	1.5	2	4	5	1
Typical current (mA)	6.5	6.5	6.5	6.5	6
Typical voltage (V)	2750	2070	2070	1910	3380

(b) The human eye is at least twice as sensitive to orange color light than for red. Discuss typical applications where it is more desirable to use the orange laser.

(c) The 1523-nm emission has potential for use in optical communications by modulating the laser beam externally. By considering the spectral line width ($\Delta\upsilon \approx 1400$ MHz), typical powers, and stability, discuss the advantages and disadvantages of using a He-Ne laser over a semiconductor diode.

4.9 The Ar-ion laser The argon-ion laser can provide powerful CW visible coherent radiation of several watts. The laser operation is achieved as follows: The Ar atoms are ionized by electron collisions in a high current electrical discharge. Further multiple collisions with electrons excite the argon ion, Ar^+, to a group of 4*p* energy levels ~35 eV above the atomic ground state as shown in Figure 4.59. Thus a population inversion forms between the 4*p* levels and the 4*s* level which is about 33.5 eV above the Ar atom ground level. Consequently, the stimulated radiation from the 4*p* levels down to the 4*s* level contains a series of wavelengths ranging from 351.1 nm to 528.7 nm. Most of the power, however, is concentrated, approximately equally, in the 488 nm and 514.5 nm emissions. The Ar^+ ion at the lower laser level (4*s*) returns to its neutral atomic ground state via a radiative decay to the Ar^+ ion ground state, followed by recombination with an electron to form the neutral atom. The Ar atom is then ready for "pumping" again.

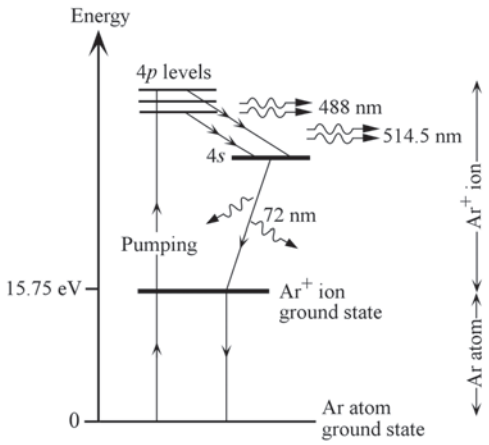


FIGURE 4.59 The Ar^+-ion laser energy levels.

(a) Calculate the energy drop involved in the excited Ar^+ ion when it is stimulated to emit the radiation at 514.5 nm.
(b) The Doppler broadened linewidth of the 514.5-nm radiation is about 3500 MHz ($\Delta\upsilon$) and is between the half-intensity points. Calculate the Doppler broadened width $\Delta\lambda$ in the wavelength.
(c) Estimate the operation temperature of the argon ion gas; give the temperature in °C.
(d) In a particular argon-ion laser the discharge tube, made of Beryllia (Beryllium Oxide), is 30 cm long and has a bore of 3 mm in diameter. When the laser is operated with a current of 40 A at 200 V DC, the total output power in the emitted radiation is 3 W. What is the efficiency of the laser?

4.10 He-Ne laser A particular commercial He-Ne laser operating at 632.8 nm has a tube that is 40 cm long. The operating temperature is 130°C.
(a) Estimate the Doppler broadened linewidth ($\Delta\lambda$) in the output spectrum.
(b) What are the mode number *m* values that satisfy the resonant cavity condition? How many modes are therefore allowed?
(c) What is the separation $\Delta\upsilon_m$ in the frequencies of the modes? What is the mode separation $\Delta\lambda_m$ in wavelength?
(d) Show that if during operation, the temperature changes the length of the cavity by δL, the wavelength of a given mode changes by $\delta\lambda_m$,

$$\delta\lambda_m = (\lambda_m/L)\delta L$$

Given that typically a glass has a linear expansion coefficient $\alpha \approx 1 \times 10^{-6}\,K^{-1}$, calculate the change $\delta\lambda_m$ in the output wavelength (due to one particular mode) as the tube warms up from 20°C to 130°C, and also per degree change in the operating temperature. Note that $\delta L/L = \alpha\delta T$, and $L' = L\left[1 + \alpha(T' - T)\right]$. Change in mode wavelength $\delta\lambda_m$ with the change δL in the cavity length *L* is called **mode sweeping**.
(e) How do the mode separations $\Delta\upsilon_m$ and $\Delta\lambda_m$ change as the tube warms up from 20°C to 130°C during operation?
(f) How can you increase the output intensity from the He-Ne laser?

4.11 Doppler broadening Consider a He-Ne laser with Doppler broadened linewidth of 2000 MHz. What should be the length of the resonator cavity so that a single mode is found to oscillate? You may assume that there is an oscillating mode at the line centre.

4.12 Einstein coefficients and critical photon concentration $\rho(\upsilon)$ is the energy of the electromagnetic radiation per unit volume per unit frequency due to photons with energy $h\upsilon = E_2 - E_1$. Suppose that there are N_{ph} photons per unit volume. Each has an energy $h\upsilon$. The frequency range of emission is $\Delta\upsilon$. Then, $\rho(\upsilon) = (N_{\text{ph}}h\upsilon)/\Delta\upsilon$. In case of an Ar-ion laser system with the emission wavelength 488 nm and the linewidth in the output spectrum is 6×10^9 Hz (between half intensity points), estimate the photon concentration necessary to achieve more stimulated emission than spontaneous emission.

4.13 Photon concentration in a gas laser The Ar-ion laser has a strong lasing emission at 488 nm. The laser tube is 1 m in length, and the bore diameter is 3 mm. The output power is 1 W. Assume that most of the output power is in the 488-nm emission. Assume that the tube end has a transmittance T of 0.1. Calculate the photon output flow (number of lasing photons emitted from the tube per unit time), photon flux (number of lasing photons emitted per unit area per unit time), and estimate the order of magnitude of the steady state photon concentration (at 488 nm) in the tube (assume that the gas refractive index is approximately 1).

4.14 He-Ne Laser and the photon cavity lifetime A He-Ne laser operating at 632.8 nm has a tube that is 40 cm long and end mirrors with reflectances 99.0% and 99.9%. The tube diameter is 0.80 mm and the output power is 1 mW. The linewidth $\Delta\upsilon$ is 1 GHz and the loss coefficient α_s is $0.06\,\text{m}^{-1}$. The spontaneous emission lifetime is 125 ns. (a) Calculate the threshold gain. (b) Calculate the threshold population inversion. (c) Calculate the photon cavity lifetime. (d) Calculate the photon concentration in the cavity.

4.15 Threshold population Nd:YAG laser The emission wavelength of a Nd:YAG laser is 1064 nm, the optical gain linewidth is about 100 GHz, the spontaneous lifetime (τ_{sp}) of the upper laser level is 200 μs. The loss coefficient as is $0.2\,\text{m}^{-1}$. A YAG rod of 10 cm in length and has its ends coated to achieve reflectances of 99.9% and 92% (emission end). What is the threshold concentration of pumped Nd-ions (Nd^{3+})?

4.16 Threshold gain and population inversion in an Ar-ion laser The emission wavelength of an Ar-ion gas laser is 488 nm. The tube length $L = 80\,\text{cm}$, tube mirror reflectances are approximately 99.0% and 96.0%. The linewidth $\Delta\upsilon = 2\,\text{GHz}$, the loss coefficient is $\alpha_s \approx 0.15\,\text{m}^{-1}$, spontaneous decay time constant $\tau_{\text{sp}} = 1/A_{21} \approx 10\,\text{ns}$, $n \approx 1$. Estimate the threshold population inversion.

4.17 Fabry–Perot optical resonator in gas lasers An idealized He-Ne laser optical cavity of length 50 cm has 99% reflectance $R_1 = R_2$ at both ends. Find out the separation $\Delta\lambda$ of the modes in wavelength and the spectral width $\Delta\lambda$ for this laser.

4.18 Homogeneous broadening

(a) Consider the emission of radiation from an isolated atom in which an electron transits down from E_2 to E_1. Suppose that the lifetime of the atom at E_2 is τ_2. We will assume that the lifetime of the state E_1 is very long and neglect it. The emitted radiation's optical field is not simply sinusoidal oscillations, sin ($\omega_o t$) type, that would go on forever since we know this is not possible. The optical field's amplitude is decayed exponentially to give the radiation a finite temporal length, which results in a finite spectral width. Suppose that the field follows $E(t) = E_o \exp(-t/2\tau_2) \times \cos(\omega_o t)$ for all $t > 0$, and zero for $t < 0$. In practice, $1/\tau_2 \ll \omega_o$. The Fourier transform (FT) $E_{\text{FT}}(\omega)$ of $E(t)$ is approximately given by

$$E_{\text{FT}}(\omega) = \text{FT of } E(t) \approx \frac{E_o/4\pi}{(2\tau_2)^{-1} + j(\omega - \omega_o)}$$

where we neglected the additional small term that has $j(\omega + \omega_o)$ in the denominator. Show that the light intensity is maximum at $\omega = \omega_o$ and that the spectrum has an FWHM width $\Delta\omega_{\text{FWHM}} = 1/\tau_2$. What is your conclusion? (The spectrum is called a *Lorentzian lineshape*.)

(b) Consider the two-energy-level system shown in Figure 4.5. The rate equation for the change in the concentration N_2 of atoms at E_2 from Eqs. (4.2.1) and (4.2.2) is

$$dN_2/dt = -A_{21}N_2 - B_{21}\rho(\upsilon)N_2 - B_{12}\rho(\upsilon)N_1$$

Since $E_2 > E_1$, we can take the equilibrium (unpumped) value of N_2 as zero. Suppose we suddenly pump the atoms to the energy level E_2 and we assume the radiation density $\rho(\upsilon)$ remains small. From the rate equation, what is the time evolution of $N_2(t)$? What does A_{21} represent? What is the time evolution of the total emitted optical power? How do your results relate to part (a)? What is your conclusion?

4.19 Homogeneous and inhomogeneous broadening The expressions for homogeneous and inhomogeneous broadening in photonics are given by the *Lorentzian* and *Gaussian* lineshape functions, *i.e.,*

$$g_L(\upsilon) = \frac{\gamma/\pi}{(\upsilon - \upsilon_o)^2 + \gamma^2} \quad \text{and} \quad g_G(\upsilon) = \frac{1}{\pi^{1/2}\Gamma}\exp\left[-(\upsilon - \upsilon_o)^2/\Gamma^2\right]$$

such as in the **Bayer color image sensor** shown in Figure 5.34 (a). In **three CCD imaging**, as shown in Figure 5.34 (b), a trichroic prism assembly is used to separate the light from the object into its red, green, and blue components, and three different CCDs are used to detect each component. A third technique uses different absorption depths for red, green, and blue light in silicon. The red, green, and blue light have different penetration depths into silicon so that, in principle, we can use photodiodes at different depths to distinguish between the colors. The Bayer filtering technique is the most common among color image sensors.

B. Active Matrix Array and CMOS Image Sensors

Many image sensors, as well as various flat panel TV technologies, make use of active matrix arrays (AMAs). An **active matrix array** is a two-dimensional array of pixels in which each pixel has a thin film transistor (a field effects transistor that has been fabricated by using thin film technologies) that can be externally addressed to read a signal from a sensor located at that pixel. Depending on the application, an AMA can have few pixels or millions of pixels. The TFT AMA technology was pioneered by Peter Brody using CdSe TFTs in early 1970s. A simplified diagram is shown in Figure 5.35 (a) for the array and in (b) for each pixel. Each pixel has a photodiode and a capacitor C_{px}. When a pixel receives light from a particular point on the illuminated object, the photodiode generates a current I_{signal} which charges the pixel capacitor C_{px}. The signal is the charge Q_{signal} stored on C_{px}. The array of pixels therefore has the image stored as charges on pixel capacitances. All we have to do is read the charges out.

Each pixel is identical with its TFT gate connected to a particular address line and the source to a particular data line as shown in Figure 5.35 (b). When row *a* is addressed, all pixels and hence all signals on that row are read out, that is, the data are read out onto the data lines as *parallel data*, which can be converted into serial data. We can scan the whole image row by row (line by line) by starting from the top row and sequentially activating one row after another. Thus, each time we activate and access a row, we extract the signals (Q_{signal} on each C_{px}) on that row as parallel data.

A **CMOS image sensor** is basically an AMA in which each pixel has a photodiode or a photogate and one or more CMOS transistors to read and amplify the electrical signal that has been

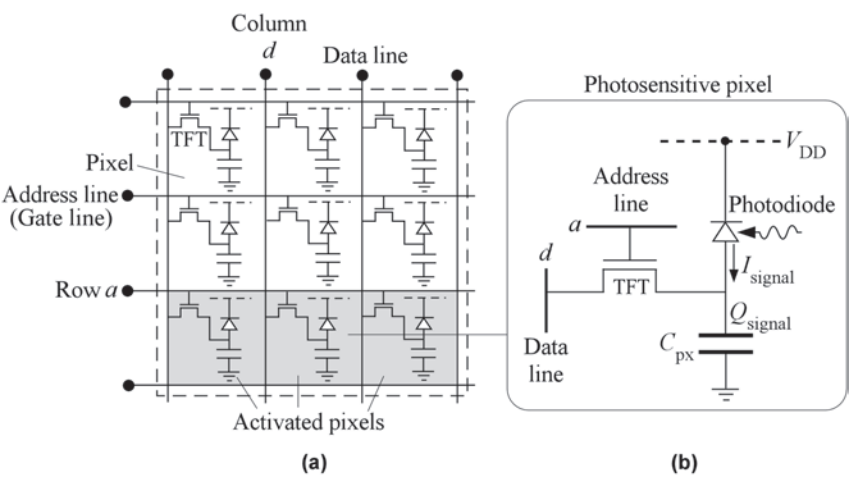


FIGURE 5.35 (a) An active matrix array (AMA). (b) A basic photosensitive pixel structure for detecting the photons arriving at the pixel defined by row *a* and column *d*.

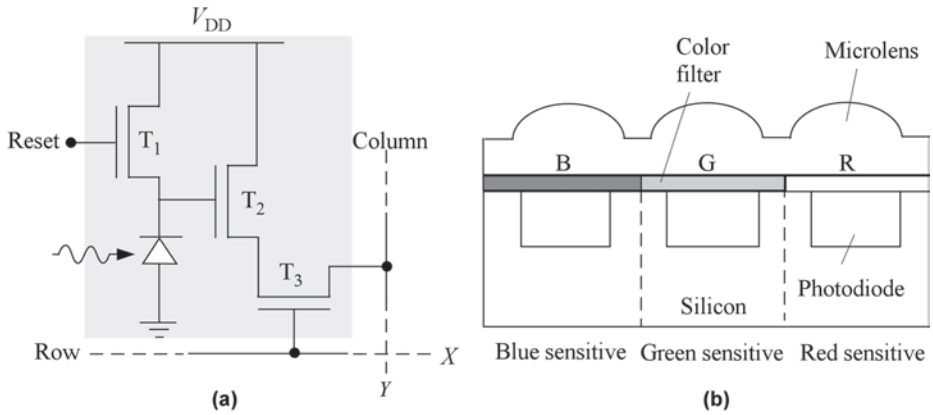
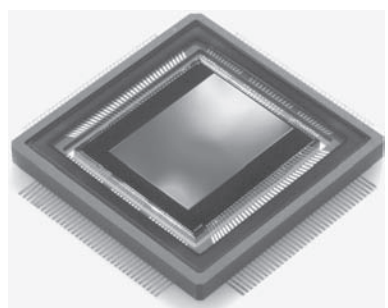


FIGURE 5.36 (a) The pixel architecture in a CMOS image sensor. (b) A cross-section(s) of a CMOS imager with microlenses and color filters (B = blue, G = green, R = red) for color imaging.

generated by the light incident on this pixel. The pixel read-out operation is essentially the same as that in the AMA. The pixel architecture, however, can change from application to application.

CMOS is a well-established silicon fabrication technology, which means that CMOS image sensors can be fabricated relatively inexpensively. If the pixel is active with gain, then the imager is called an **active pixel image sensor**, or **active pixel sensor**. In a **passive pixel sensor** there is only a switching transistor to read the charge out when the transistor switch is addressed. Nearly all modern CMOS imagers are active pixel sensors. One possible active pixel architecture using a photodiode is shown in Figure 5.36 (a). T_1 is the reset transistor, T_2 works as a source follower (*i.e.*, a buffer), and T_3 is a pixel switch transistor. With T_1 off, the photocurrent charges the self-capacitance of the photodiode to a certain voltage. When the row *X* receives a signal, T_3 is turned on, and the signal voltage on the photodiode is transferred through the buffer T_2 to the column *Y*. The pixel is then reset, and T_1 connects the photodiode to V_{DD} to clear the accumulated charge.



Four Megapixel CMOS image sensor. *(Courtesy of Teledyne-DALSA.)*

CMOS imagers have a number of distinct advantages. Standard and well-established CMOS fabrication, and hence lower cost, mean on-chip integration such as a **camera on a chip**. A camera on a chip has a microlens at each pixel—a red, green, or blue filter at each pixel for color imaging—as illustrated in Figure 5.36 (b), analog signal processing after the read-out of the image, and analog-to-digital conversion. Active pixel architecture can include pixel amplification as well to improve the overall efficiency. CMOS low power consumption leads to a longer battery life.

C. Charge-Coupled Devices

Charge-coupled devices (CCDs) are commonly used as image sensors in professional and consumer television cameras and camcorders, and as image sensors in digital still cameras. However, the term CCD in general does not imply an image sensor but a chip that is able to store and transfer signals in the form of charge. The CCD chip is an integrated circuit made from crystalline silicon and has a large number of pixels (each is a detector element); for example, a 2.5 cm × 2.5 cm CCD chip may have 1024 × 1024 or 2048 × 2048 pixels on its surface. The basic pixel structure is a MOS (metal-oxide-semiconductor) or a MIS (metal-insulator-semiconductor) device as shown in Figure 5.37. Notice that the structure is based on a *p*-type Si, an oxide layer, and a metal electrode, which is usually transparent. There is a depletion region inside the *p*-type semiconductor. The EHPs are generated inside the depletion region by illumination either from the top surface or from the backside as shown in Figure 5.37. In back-thinned CCD, light enters the depletion region not from the gate side but from the "substrate" side, which has been thinned to allow the light to pass through.

When a positive voltage $+V$ is applied to the gate V_G in Figure 5.37 ($V_G = +V$), the photogenerated electrons in the depletion region are collected in a layer near the interface as shown in Figure 5.37. (With no gate voltage, photogenerated electrons and holes disappear by recombination.) These electrons are trapped inside a potential energy well, introduced by $+V$ on the gate. Their total charge is proportional to the total light exposure. This charge constitutes the electrical signal. The objective is to read all these charges stored at the illuminated pixels.

In a **three-phase CCD** read-out, there are three line voltages V_1, V_2, V_3 to which the gates are connected in an alternating fashion: G_1 to V_1, G_2 to V_2, and G_3 to V_3, G_4 to V_1 again and so on as shown in Figure 5.38. V_1, V_2, and V_3 are appropriately clocked to shift the charges from pixel to pixel to a register located at the end of the chip. If initially (time $t = t_1$) $V_1 = +V$ and $V_2 = V_3 = 0$, then photogenerated charges will be stored under G_1, G_4, *etc.* as Q_a, Q_b, *etc.* Later ($t = t_2$) we can make $V_1 = +V$, $V_2 = +V$, and $V_3 = 0$. The charge Q_a is shared between the wells under G_1 and G_2. Even later ($t = t_3$), we can bring V_1 down to zero, the charge Q_a must go into the available potential well, which is under G_2. Thus, by toggling gate voltages, Q_a has been shifted from G_1 to G_2, and similarly Q_b from G_4 to G_5 and so on. The charges are therefore clocked progressively along the gates, from pixel to pixel, left to right, until they reach the end of the array where there is a register. The CCD read-out therefore functions like a *shift register* in that clock pulses shift the information along the chain; they are often termed *CCD shift registers*.

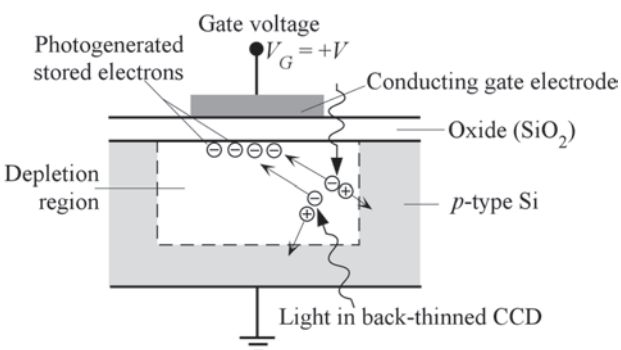
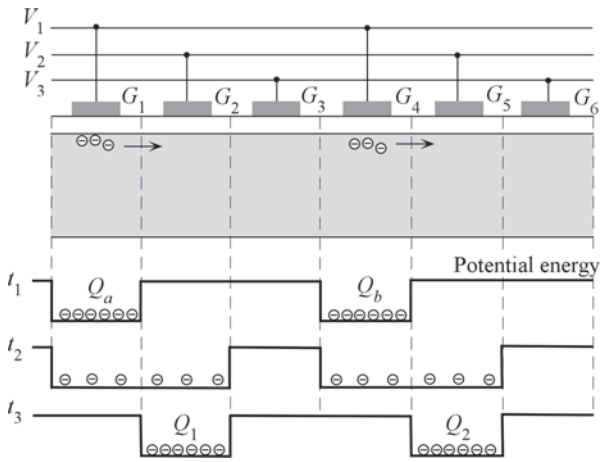


FIGURE 5.37 One element of a CCD imaging sensor, which is a MOS (metal-oxide-semiconductor) device. The absorbed photon actually creates an electron and a hole. The hole simply drifts to the *p*-side.



Various CCD chips. *(Courtesy of Teledyne-DALSA.)*



Time	V_1	V_2	V_3
t_1	$+V$	0	0
t_2	$+V$	$+V$	0
t_3	0	$+V$	0

FIGURE 5.38 Transfer of charge from one well to another by clocking the gate voltages. The table shows the gate voltage sequences in a three-phase CCD.

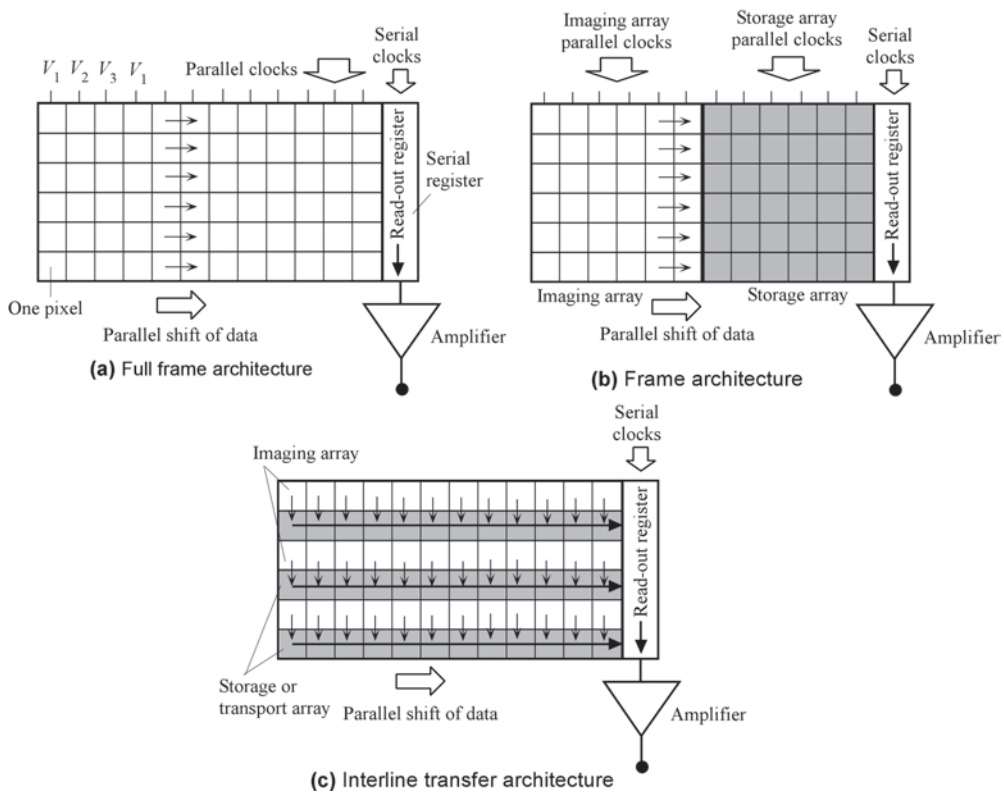


FIGURE 5.39 Three CCD architectures: (a) Full frame architecture. (b) Frame architecture. (c) Interline transfer architecture.

There are three main types of CCD read-out techniques and hence CCD architectures called full frame, frame, and interline transfer architectures. In a **full frame architecture,** the sensor consists of parallel CCD shift registers, a serial CCD shift register as a *read-out register,* and an amplifier as shown in Figure 5.39 (a). The columns of data are fed into the serial register starting soon after the projection of the image onto the array, as the clock signals start transferring the charges pixel to pixel toward the read-out register. A short build-up or exposure time is necessary to build up the charges in the imaging arrays. The process continues until all the columns of data have been transferred. This architecture has the highest resolution and is the densest.

The **frame architecture**, as shown in figure 5.39 (b), is similar to the full frame except that half the array is used as data storage to increase the speed. The storage array is blocked from light. The idea is to shift the data away from the image sensors quickly so that they can capture the next image soon; these have higher frame rates at the expense of resolution.

The **interline transfer architecture** uses every other row of the CCD array for *storage* and *transport* right next to the imaging array as illustrated in figure 5.39 (c). The transport array is blocked from light, and the data from the imaging array are transferred in parallel to the transport array after the image has been built-up in the imaging array. The data on the transport array are shifted, just as in the normal CCD.

Additional Topics

5.14 PHOTOVOLTAIC DEVICES: SOLAR CELLS

A. Basic Principles

A **photovoltaic device** or **a solar cell** converts the incident radiation energy into electrical energy.[19] Incident photons are absorbed to photogenerate charge carriers, which then pass through an external load to do electrical work. Photovoltaic devices may be metal–semiconductor Schottky junctions, *pn* junctions, or *pin* devices. For example, a crystalline Si *pn* junction solar cell may have a thin *n*-type semiconductor layer on a thick *p*-type substrate as shown in Figure 5.40 (a). The electrodes attached to the *n*-side must allow the light to enter the device and at the same time result in a small series resistance. They are deposited on to *n*-side to form an array of **finger electrodes** on the surface as illustrated in Figure 5.40 (a). There is a thin film antireflection (AR) coating (one or two layers) on the surface outside the electrodes. The AR coating significantly reduces the reflectance of the bare semiconductor–air interface.

The *n*-side of the *pn* junction is very narrow to allow most of the photons to be absorbed within the depletion region (W) and within the neutral *p*-side. The photogeneration of electron–hole pairs occurs mainly in these regions. In Si, the electron diffusion length L_e is longer than the hole diffusion length L_h, which is the reason for having the *p*-layer as the main photon absorbing layer. EHP photogeneration then occurs in a bigger volume defined by W and L_e in the *p*-side as indicated in Figure 5.40 (a). Remember that only those electrons in the *p*-side within L_e to the SCL can contribute the photocurrent. EHPs photogenerated in W in Figure 5.40 (a) are

[19] Although the terms *photovoltaic device* and *solar cell* are often used interchangeably, solar cells would convert solar radiation into electrical work, whereas a photovoltaic device could also be used as a photodetector, operating in the photovoltaic mode; their designs would be somewhat different.

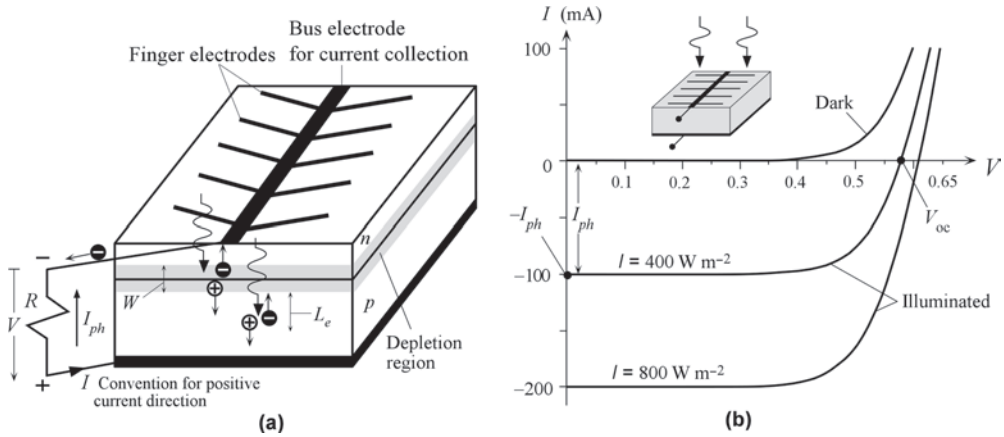


FIGURE 5.40 (a) Typical *pn* junction solar cell. Finger electrodes on the surface of a solar cell reduce the series resistance. (b) *I–V* characteristics in the dark and under illumination at intensities corresponding to 400 W m^{-2} and 800 W m^{-2}.

immediately separated by the built-in field, which drifts them apart. The electrons drift toward the *n*-side and holes toward the *p*-side, which generates a photocurrent. The excess electrons reach the neutral *n*-side, then drift around the external circuit, do work, and reach the *p*-side to recombine with the excess holes on this side. As a result, there will be a continuous external photocurrent during illumination. If the load is a short circuit, then the magnitude of the external current is simply equal to the photocurrent I_{ph} generated by the incident radiation ($I = -I_{ph}$). It is important to realize that without the internal field, it is not possible to drift apart the photogenerated EHPs and hence cause a current flow in the external circuit.

The *I–V* characteristics of a typical Si solar cell are shown in Figure 5.40 (b). The dark *I–V* is the usual forward biased *pn* junction diode equation

pn junction diode current

$$I_{\text{diode}} = I_o\left[\exp(eV/\eta k_B T) - 1\right] \tag{5.14.1}$$

where η is the diode ideality factor (not to be confused with the solar cell efficiency) which is between 1 and 2. Under illumination, the *I–V* dark characteristics are shifted down by an amount that is equal to the photocurrent I_{ph}. The photocurrent I_{ph} is proportional to the photogeneration rate and hence to the incident light intensity *I*, *i.e.,*

Photocurrent and light intensity

$$I_{ph} = KI \tag{5.14.2}$$

where *K* is a device specific constant. As can be seen from Figure 5.40 (b), doubling the light intensity has doubled I_{ph}.

The positive *I* and *V* convention and the short circuit behavior of an illuminated *pn* junction are shown in Figure 5.41 (a) and (b). The short circuit current I_{sc} is $-I_{ph}$, and is an important solar cell quantity. If the solar cell is in open circuit, there is no external current ($I = 0$) but there is a voltage, called the **open circuit voltage**, V_{oc}, across the device as in Figure 5.41 (c). This voltage corresponds to the point where the illuminated *I–V* characteristic cuts the *V*-axis in Figure 5.40 (b). The photogenerated electrons arriving on the *n*-side cannot flow in the external circuit and accumulate in the *n*-side; similarly photogenerated holes accumulate in the *p*-side. We should clearly see a net positive voltage on the *p*-side with respect to the *n*-side. The reality

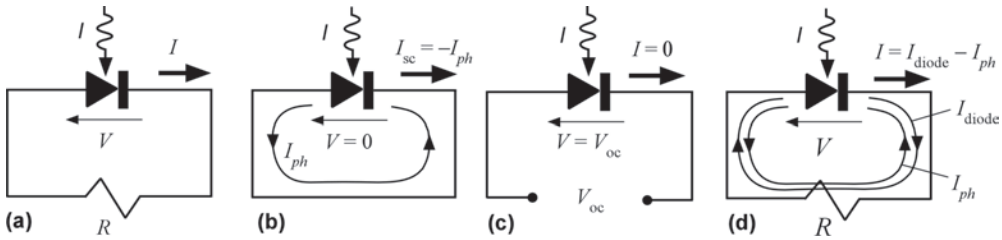


FIGURE 5.41 (a) The solar cell connected to an external load R and the convention for the definitions of positive voltage and positive current. (b) The solar cell in short circuit. The magnitude of the external current I_{sc} in the photocurrent I_{ph}. (c) Under open circuit conditions there is a voltage V_{oc} at the output terminals. (d) The solar cell driving an external load R. There is a voltage V and current I in the circuit.

is a bit more subtle. The extra electrons on the n-side neutralize some of the positive donor ions in the SCL on the n-side. Extra holes neutralize some of negative acceptors in the SCL on the p-side. The internal field becomes decreased, and hence the built-in voltage V_o becomes reduced. It is the reduction in the built-in voltage that appears as V_{oc} at the external terminals.

Suppose that there is an external load R, as shown in Figures 5.40 (a) and 5.41 (d). The external current (no longer simply I_{ph}) will flow through R and generate a voltage V across it. IV is the electric power dissipated in the external load. It should be apparent that the voltage V developed across R now *forward biases* the pn junction, generates a diode current I_{diode} in the normal way, as given by Eq. (5.14.1), and flows in the opposite direction to I_{ph}. The net current, as apparent from Figure 5.41 (d) is

$$I = -I_{ph} + I_{\text{diode}} = -I_{ph} + I_o\left[\exp(eV/\eta k_B T) - 1\right] \qquad \textbf{(5.14.3)}$$

Solar cell I–V

which represents the I–V characteristics of a solar cell. The light intensity increases I_{ph} along Eq. (5.14.2), and hence shifts the normal diode characteristics down as shown in Figure 5.40 (b).



An experimental solar cell aircraft called Helios flying over the coast of Hawaii. *(Courtesy of NASA Dryden Research Center.)*

B. Operating Current and Voltage and Fill Factor

Equation (5.14.3) gives only the I–V characteristics of the solar cell. When the solar cell is connected to a load as in Figure 5.42 (a), the load has the same voltage as the solar cell and carries the same current. But the current I through R is now in the opposite direction to the convention that current flows from high to low potential. Thus, as shown in Figure 5.42 (a)

$$I = -V/R \qquad \textbf{(5.14.4)}$$

The load line

The actual current I' and voltage V' in the circuit must satisfy both the I–V characteristics of the solar cell, Eq. (5.14.3), and that of the load, Eq. (5.14.4). We can find I' and V' by solving these two equations simultaneously or using a **load line construction**. The I–V characteristics

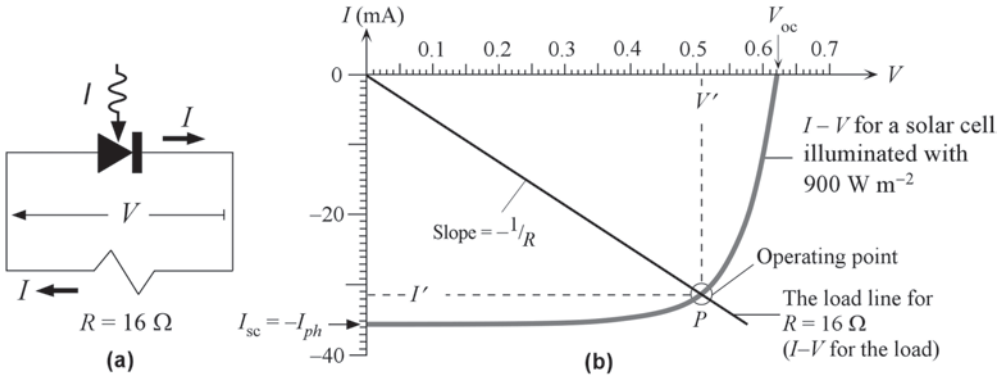


FIGURE 5.42 (a) A solar cell driving a load R and the definitions of positive current I and voltage V. (b) The load line construction for finding the operating point when a load $R_L = 15\,\Omega$ is connected across the solar cell.

of the load in Eq. (5.14.4) is a straight line with a negative slope $-1/R$. This is called the **load line** and is shown in Figure 5.42 (b) along with the I–V characteristics of the solar cell under a given intensity of illumination. The load line cuts the solar cell characteristic at P where the load and the solar cell have the same current and voltage, I' and V'. Point P therefore satisfies both Eqs. (5.14.3) and (5.14.4) and thus represents the **operating point of the circuit**.

The **power delivered** to the load is $P_{\text{out}} = I'V'$, which is the area of the rectangle bound by I- and V-axes and the dashed lines shown in Figure 5.42 (b). Maximum power is delivered to the load when this rectangular area is maximized (by changing R or the intensity of illumination), when $I' = I_m$ and $V' = V_m$. Since the maximum possible current is I_{ph} and the maximum possible voltage is V_{oc}, $I_{ph}V_{\text{oc}}$ represents the desirable goal in power delivery for a given solar cell. It therefore makes sense to compare the maximum power output[20], I_mV_m, with I_{sc} V_{oc}. The **fill factor** (FF), which is a figure of merit for the solar cell, is defined as

Definition of fill factor

$$\text{FF} = \frac{I_mV_m}{I_{\text{sc}}V_{\text{oc}}} \tag{5.14.5}$$

The FF is a measure of the closeness of the solar cell I–V curve to the rectangular shape (the ideal shape). It is clearly advantageous to have FF as close to unity as possible, but the exponential *pn* junction properties prevent this. Typically FF values are in the range 70–85% and depend on the device structure. One can vary the operating point P in Figure 5.42 (b) to examine the effects of different load lines on the rectangular area $I'V'$. A load line that cuts the solar cell I–V characteristics around the "knee" point, as in Figure 5.42 (b), corresponds to nearly the maximum power I_m,V_m point.

C. Equivalent Circuit of a Solar Cell

Practical solar cells can deviate substantially from the ideal *pn* junction solar cell behavior illustrated in Figure 5.40 (b) due to a number of reasons. Consider an illuminated solar cell driving a load resistance R_L and assume that photogeneration takes place in the depletion region. The photogenerated electrons have to traverse a surface semiconductor region to reach the nearest finger electrode as shown in Figure 5.43 (a). All these electron paths in the n-layer surface region of finger electrodes introduce an **effective series resistance** R_s into the photovoltaic circuit. If the finger electrodes are thin, then the resistance of the electrodes themselves will further

[20] One should not be too concerned with the negative sign for I_mV_m inasmuch as the diode is generating electric power during illumination.

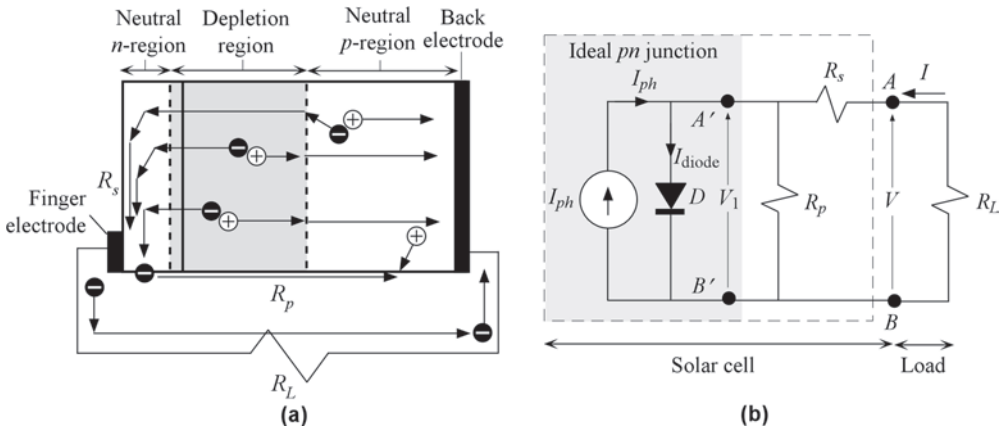


FIGURE 5.43 (a) Series and shunt resistances and various fates of photogenerated EHPs. (b) A simple equivalent circuit diagram for a solar cell. The current I into A, and V across A and B, follow the convention for positive current and voltage. A' and B' represent the ideal diode terminals.

increase R_s. There is also a series resistance due to the neutral p-region but this is generally small compared with the resistance of the electron paths to the finger electrodes.

Figure 5.43 (b) shows the equivalent circuit of a pn junction solar cell. The gray region is the ideal pn junction diode case and has the terminals A' and B'. The photogeneration process is represented by a constant current generator I_{ph}, which generates a current that is proportional to the light intensity. The flow of photogenerated carriers across the junction gives rise to a photovoltaic voltage difference V_1 across the junction, and this voltage leads to the normal diode current $I_{\text{diode}} = I_o\left[\exp(eV_1/\eta kT) - 1\right]$. This diode current I_{diode} is represented by an ideal pn junction diode D in the circuit as shown in Figure 5.43 (b). As apparent, I_{ph} and I_{diode} are in opposite directions (I_{ph} is "up" and I_{diode} is "down") so that, in open circuit, the photovoltaic voltage V_1 is such that I_{ph} and I_{diode} have the same magnitude and cancel each other. By convention, positive current I at the output terminal is normally taken to flow into the terminal, and is given by Eq. (5.14.3) (In reality, of course, the solar cell current is negative, which represents a current that is flowing out into the load.) The voltage V between A and B is the actual voltage measured, and available externally across the solar cell terminals. Notice that $V_1 = V - IR_s$ so that we can substitute for V_1 in terms of V in the diode current equation.

The equivalent circuit in Figure 5.43 (b) outside the ideal gray box contains the series resistance R_s in Figure 5.43 (a), which gives rise to a voltage drop, and therefore prevents the ideal photovoltaic voltage from developing at the output between A and B when a current is drawn. A small fraction of the photogenerated carriers can also flow through the crystal surfaces (edges of the device) or through *grain boundaries in polycrystalline devices* instead of flowing through the external load R_L as shown in Figure 5.43 (a). These effects that prevent photogenerated carriers from flowing in the external circuit can be represented by an effective **shunt** or **parallel resistance** R_p that diverts the photocurrent away from the load R_L. Typically R_p is less important than R_s in overall device behavior, unless the device is highly polycrystalline and the current component flowing through grain boundaries is not negligible. The current I flowing into A, and the voltage V across A and B (of A with respect to B) in Figure 5.43 (b) now represent the solar cell characteristics that the user measures.

The series resistance R_s can significantly deteriorate the solar cell performance as illustrated in Figure 5.44, where $R_s = 0$ is the best solar cell case. It is apparent that the available maximum output power decreases with increasing series resistance, which therefore reduces the

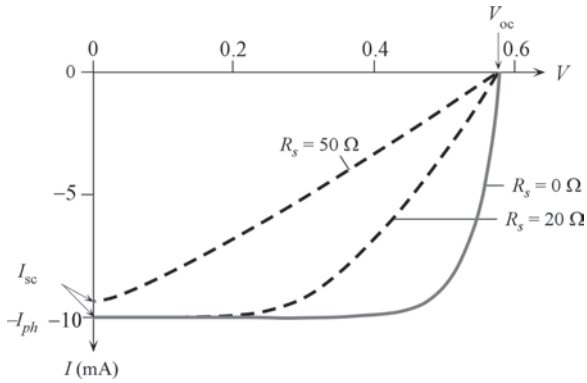


FIGURE 5.44 The effect of series resistance R_s on the *I–V* characteristics. This example is a Si *pn* junction solar cell with $\eta = 1.5$ and $I_o = 3 \times 10^{-6}$ mA. The light intensity is such that it generates $I_{ph} = 10$ mA.

cell efficiency. Notice also that, as expected, V_{oc} is unaffected by R_s, though for sufficiently large R_s, the short circuit current can become reduced. Similarly, low shunt resistance values, due to extensive defects in the material, also reduce the efficiency. The difference is that although R_s does not affect the open circuit voltage V_{oc}, low R_p leads to a reduced V_{oc}. (See Question 5.35.)

D. Solar Cell Structures and Efficiencies

Most solar cells use crystalline silicon because silicon-based semiconductor fabrication is now a mature technology that enables cost-effective devices to be manufactured. One of the most important solar cell metrics is the **efficiency**, which is defined as the maximum output power from the device per unit incident radiation power under well-defined conditions. The Sun's spectrum is different at different heights from Earth due to absorption and scattering effects in the atmosphere, which depend on the wavelength. For terrestrial use, one common standard condition for solar cell comparisons is the so-called global **AM1.5 solar spectrum**[21] that represents the radiation arriving on Earth's surface with an integrated intensity (over all wavelengths) of 1000 W m^{-2}; and the test temperature is 25°C. Table 5.6 lists the efficiencies and various solar cell parameters such as V_{oc}, J_{sc} (short circuit current density), and the fill factor.

TABLE 5.6 Characteristics of a few selected classes of solar cells, and reported efficiencies under global AM1.5 solar spectrum (1000 W m^{-2}) at 25°C

Semiconductor	V_{oc} (V)	$\lvert J_{sc} \rvert$ (mA cm^{-2})	FF (%)	Efficiency (%)	E_g (eV)
Si, single crystal (PERL)	0.707	42.7	82.8	25.0	1.11
Si, polycrystalline	0.664	38.0	80.9	20.4	1.11
Amorphous Si:H (*pin*)	0.886	16.75	67.0	10.1	~1.7
GaAs, single crystal	1.030	29.8	86.0	26.4	1.42
GaAs, thin film	1.107	29.6	84.1	27.6	1.42
InP, single crystal	0.878	29.5	85.4	22.1	1.35
GaInP/GaAs Tandem	2.488	14.22	85.6	30.3	1.95/1.42
GaInP/GaAs/Ge Tandem	2.622	14.37	85.0	32.0	1.95/1.42/0.66

(*Source:* Data have been extracted from M. A. Green, K. Emery, Y. Hishikawa, W. Warta, *Progress in Photovoltaics: Research and Applications, 18*, 346, 2010, and *19*, 84, 2011. The original tables in the latter have extensive confirmed data on a number of important solar cells and modules with references. In addition, the original tables also list the uncertainties and errors involved in the reported efficiency values.)

[21] AM1.5 means Air-Mass1.5. In general, the number m in AMm represents $m = \sec\theta$, where θ is the Zenith angle between the shortest path the sun rays can take and the actual path to the solar cell.

Typical Si-based solar cell efficiencies range from about 18% for polycrystalline to 22–25% in high efficiency single crystal devices that have special structures to absorb as many of the incident photons as possible. The reflection of light from the solar cell surface must be minimized to increase the device efficiency. Indeed, best Si *pn* junction solar cell efficiencies are about 24–25% for expensive single crystal PERL—Passivated Emitter Rear Locally diffused—cells.[22] The PERL and similar cells have a textured surface that is an array of "inverted pyramids" etched into the surface to capture as much of the incoming light as possible as illustrated in Figure 5.45. Normal reflections from a flat crystal surface lead to a loss of light, whereas reflections inside the pyramid allow a second or even a third chance for absorption. Further, after refraction photons would be entering the semiconductor at oblique angles, which means that they will be absorbed in the useful photogeneration volume, that is, within the electron diffusion length of the depletion layer as shown in Figure 5.45. The largest factors reducing the efficiency of a Si solar cell are the unabsorbed photons with $h\upsilon < E_g$ (long wavelengths) and short wavelength photons that are absorbed near the surface. Both of these factors are improved by the use of multifunction heterostructure.

Research over the years has led to the development of high efficiency heterostructure **tandem** solar cells, which are also called **multijunction solar cells.** Multijunction solar cells essentially use two or more cells in tandem, or in cascade, to increase the absorbed photons from the incident light as illustrated in Figure 5.46 (a). The first cell is made from a wider bandgap material and only absorbs photons with $h\upsilon > E_{g1}$. The second cell absorbs photons that pass the first cell and have $h\upsilon > E_{g2}$. The whole structure can be grown by using lattice matched crystalline layers on a suitable substrate, leading to a monolithic tandem cell. The two cells have to be connected, that is, allow the carriers (electrons and holes) to pass. This is done by using a highly doped very thin *pn* junction between the two cells that serves as a tunneling junction; the carriers tunnel through it. All the layers are grown by special techniques on a single substrate. One of the best efficiencies is achieved by using a three junction solar cell, which is illustrated in Figure 5.46 (b). The layers are all grown on a Ge substrate and each cell is an *np* junction. There are two very thin *pn* tunnel junctions that connect the cells in tandem as shown in Figure 5.46 (b), to allow the drifting carriers tunnel through (pass through). The top cell is $GaInP_2$ with $E_g \approx 1.95\,\text{eV}\,(\lambda_g = 0.64\,\mu\text{m})$, the second is GaAs with $E_g \approx 1.42\,\text{eV}\,(\lambda_g = 0.87\,\mu\text{m})$, and the third is Ge with $E_g \approx 0.66\,\text{eV}\,(\lambda_g = 1.9\,\mu\text{m})$. The three cells have a wide spectral range and are able to capture a very high percentage of the solar radiation. The multijunction solar cell in

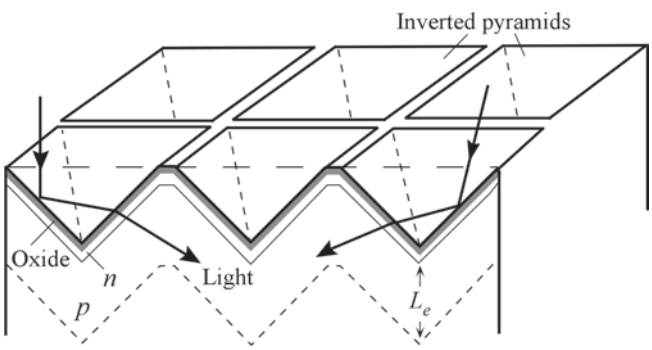


FIGURE 5.45 Inverted pyramid textured surface substantially reduces reflection losses and increases absorption probability in the device.

[22] Much of the pioneering work for high efficiency PERL solar cells was done by Martin Green and coworkers at the University of New South Wales, Australia, (J. Zhao, A. Wang, and M. A. Green, *Appl. Phys. Letts.*, *73*, 1991, 1998.)

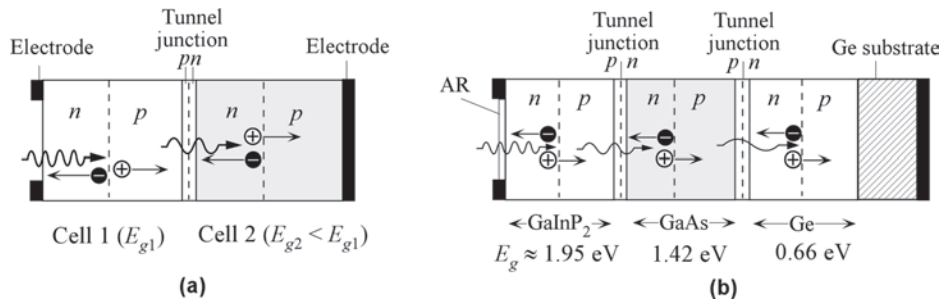


FIGURE 5.46 (a) A heterojunction solar cell that can absorb both high and low energy photons and generate a photocurrent. (b) A tandem solar cell.

Figure 5.46 (b) is commercially available (Spectrolab) with an efficiency of 29.5%. Even higher efficiencies have been reported in research labs using such multijunction heterostructures. If, in addition, light concentrators are also used, the efficiency can be further increased.

Tandem cells are also used in thin film a-Si:H (hydrogenated amorphous Si) *pin* solar cells to obtain efficiencies up to about 12%. These tandem cells are based on a-Si:H and a-SiGe:H and are easily and inexpensively fabricated in large areas. Amorphous Si:H has an E_g of about 1.8 eV. The alloying of a-Si:H with Ge to produce a-SiGe:H decreases E_g. Further, E_g of a-SiGe:H can be graded by controlling the Ge content.

EXAMPLE 5.14.1 Solar cell driving a load

Consider the solar cell driving a 16-Ω resistive load as in Figure 5.42 (b). Suppose that the cell has an area of 1 cm × 1 cm and is illuminated with light of intensity 900 W m^{-2} as in the figure. What are the current and voltage in the circuit? What is the power delivered to the load? What is the efficiency of the solar cell in this circuit? If you assume it is operating close to the maximum deliverable power, what is the FF?

Solution

The *I–V* characteristic of the load is the load line as described by Eq. (5.14.4), $I = -V/R$ with $R = 16\ \Omega$. This line is drawn in Figure 5.42 (b) with a slope $1/(16\ \Omega)$. It cuts the *I–V* characteristics of the solar cell at $I' \approx -31.5$ mA and $V' \approx 0.505$ V, which are the current and voltage in the photovoltaic circuit of Figure 5.42 (b). In fact, from Eq. (5.14.4), V'/I' gives $-16\ \Omega$ as expected. The power delivered to the load is

$$P_{\text{out}} = |I'V'| = (31.5 \times 10^{-3}\ \text{A})(0.505\ \text{V}) = 0.0159\ \text{W} \quad \text{or} \quad 15.9\ \text{mW}$$

This is *not* necessarily the maximum power available from the solar cell. The input sunlight power is

$$P_{\text{in}} = (\text{Light intensity})(\text{Surface area}) = (900\ \text{W m}^{-2})(0.01\ \text{m})^2 = 0.090\ \text{W}$$

The efficiency is

$$\text{Efficiency} = 100 \times (P_{\text{out}}/P_{\text{in}}) = 100(15.9\ \text{mW}/90\ \text{mW}) = 17.7\%$$

This will increase if the load is adjusted to extract the maximum power from the solar cell but the increase will be small as the rectangular area $I'V'$ in Figure 5.42 in it is already close to the maximum. Assuming that $|I'V'|$ is roughly the maximum power available (maximum area for the rectangle $I'V'$), then $I_m \approx I' \approx -31.5$ mA and $V_m \approx V' \approx 0.505$ V. For the solar cell in Figure 5.42 (b), $I_{\text{sc}} = -35.5$ mA and $V_{\text{oc}} = 0.62$ V. Then

$$\text{FF} = I_m V_m / I_{\text{sc}} V_{\text{oc}} \approx (-31.5\ \text{mA})(0.505\ \text{V})/(-35.5\ \text{mA})(0.62\ \text{V}) = 0.72 \quad \text{or} \quad 72\%$$

EXAMPLE 5.14.2 Open circuit voltage and short circuit current

A solar cell under an illumination of 500 W m^{-2} has a short circuit current I_{sc} of −16 mA and an open circuit output voltage V_{oc} of 0.50 V. What are the short circuit current and open circuit voltages when the light intensity is doubled? Assume $\eta = 1$.

Solution

The general *I–V* characteristics under illumination is given by Eq. (5.14.3). The short circuit current corresponds to the photocurrent so that, from Eq. (5.14.2), at double the intensity, the photocurrent is

$$I_{ph2} = \left(\frac{I_2}{I_1}\right) I_{ph1} = (16\text{ mA})(1000/500) = 32\text{ mA}$$

Setting $I = 0$ for open circuit we can obtain the open circuit voltage V_{oc},

$$I = -I_{ph} + I_o\left[\exp(eV_{oc}/\eta k_B T) - 1\right] = 0$$

Assuming that $V_{oc} \gg \eta k_B T/e$, rearranging the above equation we can find V_{oc}

$$V_{oc} = \frac{\eta k_B T}{e} \ln\left(\frac{I_{ph}}{I_o}\right) \tag{5.14.6}$$

Open circuit output voltage

In Eq. (5.14.6), the photocurrent, I_{ph}, depends on the light intensity I via $I_{ph} = KI$. At a given temperature, then the change in V_{oc} is

$$V_{oc2} - V_{oc1} = \frac{\eta k_B T}{e} \ln\left(\frac{I_{ph2}}{I_{ph1}}\right) = \frac{\eta k_B T}{e} \ln\left(\frac{I_2}{I_1}\right)$$

Assuming $\eta = 1$, the new open circuit voltage is

$$V_{oc2} = V_{oc1} + \frac{\eta k_B T}{e} \ln\left(\frac{I_2}{I_1}\right) = 0.50\text{ V} + (1)(0.0259\text{ V})\ln(2) \approx 0.52\text{ V}$$

This is a ~4% increase in V_{oc} compared with the 100% increase in illumination and the short circuit current.

Questions and Problems

5.1 Bandgap and photodetection

(a) Determine the maximum value of the energy gap which a semiconductor, used as a photoconductor, can have if it is to be sensitive to light of wavelength 1550 nm commonly used in optical communication network.

(b) A photodetector whose area is 4×10^{-2} cm^2 is irradiated with light of wavelength 1300 nm, whose intensity is 3 mW cm^{-2}. Assuming that each photon generates one electron–hole pair, calculate the number of pairs generated per second.

(c) From the known energy gap of the semiconductor InGaAsP ($E_g = 1.24$ eV), calculate the primary wavelength of photons emitted from this crystal as a result of electron–hole recombination. Is this wavelength in the visible range?

(d) Will a silicon photodetector be sensitive to the radiation from an InGaAsP laser? Why?

5.2 Absorption coefficient In the following questions assume that all the incident radiation onto the semiconductor surface is transmitted into the semiconductor medium; that is, there is a perfect antirelection coating on the surface of the semiconductor.

(a) If d is the thickness of a photodetector material, I_o is the intensity of the incoming radiation, show that the number of photons absorbed per unit volume of sample is

$$n_{ph} = \frac{I_o\left[1 - \exp(-\alpha d)\right]}{dh\nu}$$

(b) What is the thickness of a Ge and $In_{0.53}Ga_{0.47}As$ crystal layer that is needed for absorbing 90% of the incident radiation at 1.5 μm? (Use Figure 5.5)

(c) Suppose that each absorbed photon liberates one electron (or electron hole pair) in a unity quantum efficiency photodetector and that the photogenerated electrons are immediately collected. Thus, the rate of charge collection is limited by rate of photon generation. What is the external photocurrent density for the photodetectors in (b) if the incident radiation is 100 μW mm^{-2}?

5.3 Absorption coefficient and device design An optoelectronic device engineer has designed a *pn* junction photodiode that has a depletion width of 2 μm. The semiconductor on the light-receiving side is heavily doped (*e.g.*, p^+), the neutral region is very narrow, and can be neglected. If the photogeneration is to occur in the depletion region, what are approximate corresponding wavelengths for the semiconductors in Figure 5.5?

5.4 Absorption coefficient and quantum efficiency Consider the Si *pn* junction photodiode shown in Figure 5.8. Suppose that the incident optical power is P_o. The incident number of photons per unit time is $P_o/h\upsilon$. Of these, a fraction T is transmitted. The number of photons flowing per second along the device is

$$\frac{dN_{\text{ph}}}{dt} = \frac{TP_o}{h\upsilon}\exp(-\alpha x)$$

where x is measured from the illuminated surface of the semiconductor. Show that the photocurrent generated by the photodiode is given by

$$I_{ph} = \frac{e\eta_i TP_o}{h\upsilon}\left\{\exp\left[-\alpha(\ell_p - L_e)\right] - \exp\left[-\alpha(\ell_p - L_e + W + L_h)\right]\right\}$$

where η_i is the internal quantum efficiency and the other symbols are defined in Figure 5.8. Consider two photodiodes, one p^+n and the other *pin*. Their properties are listed in Table 5.7, where the photocurrent has been calculated at 800 nm. Suppose that the incident optical power is 1 μW, and the antireflection coating on the semiconductor provides $T \approx 1$. Find the photocurrent I_{ph} and hence the responsivity of the two photodiodes at 900 nm and 500 nm. What is your conclusion?

TABLE 5.7 Properties of a p^+n and *pin* photodiode

	ℓ_p (μm)	L_e (μm)	W (μm)	L_h (μm)	α (m^{-1}) (at 800 nm)	I_{ph} (μA)
p^+n	0.5	0.1	1	10	1×10^5	0.45
pin	0.5	0.1	30	0.1	1×10^5	0.59

Note: $\eta_i = 1$, $T = 1$ and $P_o = 1$ μW were used.

5.5 Ge Photodiode Consider a commercial Ge *pn* junction photodiode which has the responsivity shown in Figure 5.47. Its photosensitive area is 0.008 mm^2. It is used under a reverse bias of 10 V when the dark current is 0.3 μA and the junction capacitance is 4 pF. The rise time of the photodiode is 0.5 ns.

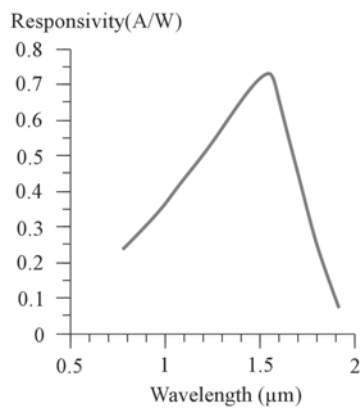


FIGURE 5.47 The responsivity of a commercial Ge *pn* junction photodiode.

(a) Calculate its quantum efficiency at 850 nm, 1300 nm, and 1550 nm.
(b) What is the light intensity of light at 1.55 μm that gives a photocurrent equal to the dark current?
(c) What would be the effect of lowering the temperature on the responsivity curve?
(d) Given that the dark current is in the range of microamperes, what would be the advantage in reducing the temperature?
(e) Suppose that the photodiode is used with a 100-Ω resistance to sample the photocurrent. What limits the speed of response?

5.6 Si *pin* Photodiodes Consider two commercial Si *pin* photodiodes, type A and type B, both classified as fast *pin* photodiodes. They have the responsivity curves shown in Figure 5.48. Differences in the responsivity are due to the *pin* device structure. The photosensitive area is 0.125 mm^2 (0.4 mm in diameter).

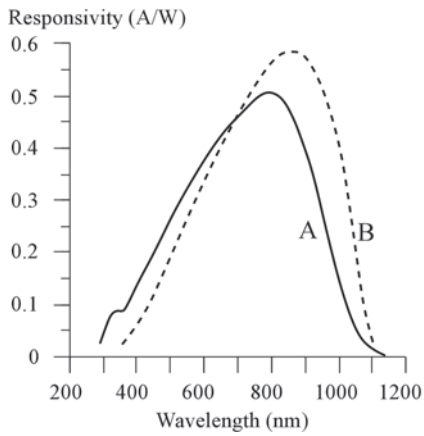


FIGURE 5.48 The responsivity of two commercial Si *pin* photodiodes.

(a) Calculate the photocurrent from each when they are illuminated with blue light of wavelength 450 nm and light intensity 1 μW cm^{-2}. What is the QE of each device?
(b) Calculate the photocurrent from each when they are illuminated with red light of wavelength 700 nm and light intensity 1 μW cm^{-2}. What is the QE of each device?
(c) Calculate the photocurrent from each when they are illuminated with infrared light of wavelength 1000 nm and light intensity 1 μW cm^{-2}. What is the QE of each device?
(d) What is your conclusion?

5.7 InGaAs *pin* Photodiodes Consider a commercial InGaAs *pin* photodiode whose responsivity is shown in Figure 5.49. Its dark current is 5 nA.

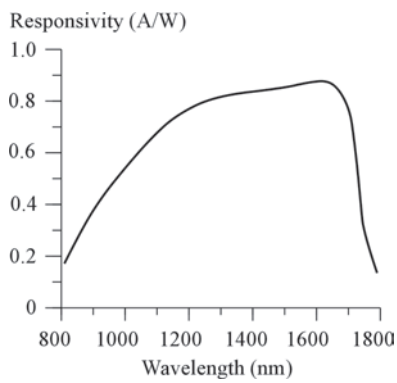


FIGURE 5.49 The responsivity of an InGaAs pin photodiode.

(a) What optical power at a wavelength of 1.55 μm would give a photocurrent that is twice the dark current? What is the QE of the photodetector at 1.55 μm?
(b) What would be the photocurrent if the incident power in (a) was at 1.3 μm? What is the QE at 1.3 μm operation?

5.8 InAs Photodiode Figure 5.50 shows the responsivity of an InAs *pn* junction photodiode for use in the infrared. It is usually cooled under normal operation.

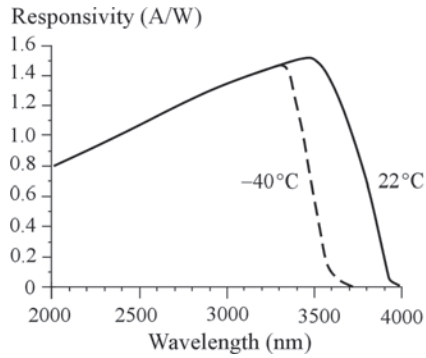


FIGURE 5.50 Responsivity of an InAs photodiode.

(a) What is the wavelength and the quantum efficiency at the peak responsivity at −40°C?
(b) The diode area that is involved in the detection has a diameter of 0.25 mm. What is the intensity of light that corresponds to a photocurrent of 50 nA at peak responsivity?
(c) Why does the cutoff wavelength become shorter at lower temperatures?

5.9 Photocurrent *pin* photodiode Consider a Si *pin* photodiode with a depletion layer width of 30 μm and diameter of 300 μm. The absorption coefficient at 800 nm is 10^3 cm^{-1}.

Find the quantum efficiency of this photodiode, assuming perfect AR coating. If the load resistance is 10 kΩ, obtain the RC time constant. What would happen if there is no AR coating so that $T = 0.70$?

5.10 Shockley–Ramo theorem Consider a *pin* photodiode with an *i*-layer width *W*. A very short pulse of light is absorbed just inside the depletion region on the p^+-side (Figure 5.9). The photogenerated electrons drift through the *i*-layer with a velocity given in Figure 5.10. What is the current generated by this drift? How long does it last? Suppose that *W* is 30 μm, and assume that the quantum efficiency is 0.80. A voltage of 20 V is applied to reverse bias the *pin* detector. The light pulse from a femtosecond laser operating at 515 nm is used for photoexcitation, and the light pulse energy is 37.5 fJ. Assume all of this energy is absorbed very close to the depletion region and p^+-layer boundary, *i.e.* electrons drift across *W* and constitute the photocurrent. Calculate the transient photocurrent. How long does it last?

5.11 QE and Responsivity A photodiode has a quantum efficiency of 70% when photons of energy 1.5×10^{-19} J are incident upon it. What is the operating wavelength of this photodiode? Find out the incident optical power required to obtain a photocurrent of 2.5 μA with the above operating wavelength. What would be the responsivity when the QE is 100%?

5.12 Si *pin* photodiode speed Consider Si *pin* photodiodes with a p^+-layer of thickness 1.0 μm, and an *i*-Si layer of width 7.5 μm. It is reverse biased with a voltage of 25 V.

(a) What is the speed of response due to bulk absorption? What wavelengths would lead to this type of speed of response?
(b) What is the speed of response due to absorption near the surface? What wavelengths would lead to this type of speed of response?

5.13 Transient photocurrents in a *pin* photodiode Consider a reverse biased Si *pin* photodiode as shown in Figure 5.51. It is appropriately reverse biased so that the field in the depletion region (*i*-Si layer), $E = V_r/W$, is the saturation field. Thus, photogenerated electrons and holes in this layer drift at saturation velocities v_{de} and v_{dh}. Assume that the field *E* is uniform and that the thickness of the p^+ is negligible. A very short light pulse (infinitesimally short) photogenerates EHPs in the depletion layer as shown in Figure 5.51, which results in an exponentially decaying EHP concentrations across *W*. Figure 5.51 shows the photogenerated electron concentration at time $t = 0$ and also at a later time *t*, when the electrons have drifted a distance $\Delta x = v_{de}t$. Those that reach the back electrode *B* become collected. The electron distribution shifts at a constant velocity until the

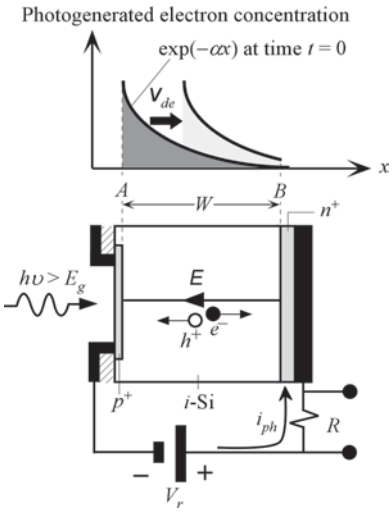


FIGURE 5.51 An infinitesimally short light pulse is absorbed throughout the depletion layer of a reverse-biased ideal *pin* photodetector, and creates an EHP concentration that decays exponentially. The p^+- and n^+-layers are assumed to be very thin.

initial electrons at A reach B, which represents the longest transit time $\tau_e = W/v_{de}$. Similar arguments apply to holes, but they drift in the opposite direction and their transit time $\tau_h = W/v_{dh}$ where v_{dh} is their saturation velocity. The photocurrent density at any instant is

$$j_{ph} = j_e(t) + j_h(t) = eN_e v_{de} + eN_h v_{dh}$$

where N_e and N_h are the overall electron and hole concentration in the sample at time t. Assume for convenience that the cross-sectional area $A = 1$ below (derivations are not affected as we are interested in the photocurrent current densities).

(a) Sketch the hole distribution at time $\tau_h > t > 0$ where τ_h = hole drift time $= W/v_{dh}$.

(b) The electron concentration distribution $n(x)$ at time t corresponds to that at $t = 0$ shifted by $v_{de}t$. Thus the total electrons in W is proportional to integrating this distribution $n(x)$ from A at $x = v_{de}t$ to B at $x = W$. Given $n(x) = n_o \exp(-\alpha x)$ at $t = 0$, where n_o is the electron concentration at $x = 0$ at $t = 0$ we have

$$\text{Total number electrons at time } t = \int_{v_{de}t}^{W} n_o \exp\left[-\alpha(x - v_{de}t)\right] dx$$

and

$$N_e(t) = \frac{\text{Total number electrons at time } t}{\text{Volume}}$$

Then

$$N_e(t) = \frac{1}{W}\int_{v_{de}t}^{W} n_o \exp\left[-\alpha(x - v_{de}t)\right] dx$$

$$= \frac{n_o}{W\alpha}\left\{1 - \exp\left[-\alpha W\left(1 - \frac{t}{\tau_e}\right)\right]\right\}$$

$N_e(0)$ is the initial overall electron concentration at time $t = 0$, that is,

$$N_e(0) = \frac{1}{W}\int_0^W n_o \exp(-\alpha x) dx = \frac{n_o}{W\alpha}\left[1 - \exp(-\alpha W)\right]$$

We note that n_o depends on the intensity I of the light pulse so that $n_o \propto I$. Show that for holes

$$N_h(t) = \frac{n_o \exp(-\alpha W)}{W\alpha}\left\{\exp\left[\alpha W\left(1 - \frac{t}{\tau_h}\right)\right] - 1\right\}$$

(c) Given $W = 40\ \mu m$, $\alpha = 5 \times 10^4\ m^{-1}$, $v_{de} = 10^5\ m\ s^{-1}$, $v_{dh} = 0.8 \times 10^5\ m\ s^{-1}$, $n_o = 10^{13}\ cm^{-3}$, calculate the electron and hole transit time, sketch the photocurrent densities $j_e(t)$ and $j_h(t)$ and hence $j_{ph}(t)$ as a function of time, and calculate the initial photocurrent. What is your conclusion?

5.14 Fiber attenuation and InGaAs *pin* Photodiode Consider the commercial InGaAs *pin* photodiode whose responsivity is shown in Figure 5.49. This is used in a receiver circuit that needs a minimum of 6 nA photocurrent for a discernible output signal (acceptable signal to noise ratio for the customer). Suppose that the InGaAs *pin* PD is used at 1.3-μm operation with a single mode fiber whose attenuation is 0.4 dB km^{-1}. If the laser diode emitter can launch at most 3 mW of power into the fiber, what is the maximum distance for the communication without a repeater?

5.15 Multiplication in APDs Consider the results of multiplication vs. reverse bias experiments on a commercial InGaAs APD summarized in Table 5.8. Plot $(1 - 1/M)$ vs. V_r and hence find m and V_{br} in Eq. (5.6.2). (This can be done, for example, by using Excel and then fitting a power law trend line; or plotting on a log-log axis in which case the slope would give m.)

TABLE 5.8 Multiplication (M) vs. reverse voltage (V_r) data for an InGaAs commercial APD

V_r	30.6	33.5	36.6	39.5	41.9	43.4	44.3	45.46	45.92	46.21
M	1.78	2.16	2.74	3.74	5.34	7.21	9.0	14.2	19.5	24.8

5.16 Effect of temperature on APDs Experiments on two commercial Si APDs A and B have examined how the temperature affects the reverse bias V_r needed to keep the multiplication constant. The results are summarized in Table 5.9 in terms of V_r values at different temperatures that would keep $M = 100$ and 200 for A and 100 for B. Plot V_r vs. T (in K) for A and B. Do you agree that empirically $V_r \propto T^n$? What is n?

TABLE 5.9 The reverse bias voltage V_r for given multiplication M at different temperatures for two APDs A and B

T	−20°C	0°C	20°C	40°C	60°C	Comment
V_r at $M = 100$	125	162.5	200	236	273	APD A
V_r at $M = 100$	135	173.5	211	249	286	APD A
V_r at $M = 100$	147	150	153	155	157	APD B

5.17 Multiplication in APDs The avalanche multiplication gain M is given by

$$M = \frac{1 - k}{\exp\left[-(1 - k)\alpha_e w\right] - k}$$

Consider two cases (a) $k = 0$ and (b) $k = 0.1$. Suppose that w is 2 μm. Increasing the field will increase α_e. Plot M vs. α_e semilogarithmically for the two cases over the range $\alpha_e = 0$ to 2. What is your conclusion? (*Note*: As α_e increases with the field, k does not strictly remain the same. However, the two cases clearly distinguish two distinctly different types of behavior.)

5.18 Si APD The electron and hole ionization coefficients α_e and α_h in silicon are approximately given in Example 5.6.3. Suppose we would like the avalanche to be achieved at a certain applied field in the avalanche region (corresponding to a particular desirable voltage range). Suppose that we would like the multiplication M to be 100 when $E = 5 \times 10^5\ V\ cm^{-1}$. Estimate the width w of the avalanche region.

5.19 InP APD design For InP the impact ionization coefficients are roughly given by $\alpha_h \approx (9.2 \times 10^6) \exp(-3.44 \times 10^6/E)$ and $\alpha_h \approx (4.3 \times 10^6)\exp(-2.72 \times 10^6/E)$, where α_e and α_h are in cm^{-1}, and E is in $V\ cm^{-1}$ (from K. Takuchi *et al.*, *J. Appl. Phys.*, *59*, 476, 1986). Consider the heterojunction InGaAs-InP APD in Figure 5.16. The avalanche is initiated by holes in N-InP. We can define $k = \alpha_e/\alpha_h$ and then the multiplication $M = (1 - k)/\left\{\exp\left[-(1 - k)\alpha_h w\right] - k\right\}$. Suppose that the width of the N-layer is 1 μm and the applied field is $4.6 \times 10^5\ V\ cm^{-1}$. What is the multiplication M? If you increase the field by 2%, what is M?

5.20 Schottky junction photodiode Consider a *pn* junction and a Schottky junction photodiode. Calculate the photocurrent under an illumination of 2 μW at a wavelength of 400 nm for each, and find the responsivity given (a) the *pn* junction has $\ell_p = 0.5\ \mu m$, $L_e = 0.1\ \mu m$, $W = 2\ \mu m$, $L_h = 10\ \mu m$; and (b) the Schottky junction has $W = 3\ \mu m$. Assume that the internal QE is 100%. (*Note*: You can use the equation in Question 5.3.)

5.21 Photoconductive detector An *n*-type Si photoconductor has a length $L = 100\ \mu m$ and a cross-sectional area $A = 10^{-4}\ mm^2$. The applied bias voltage to the photoconductor is 10 V.

(a) What are the transit times, t_e and t_h, of an electron and a hole across L? What is the photoconductive gain?

(b) It should be apparent that as electrons are much faster than holes, a photogenerated electron leaves the photoconductor very quickly. This leaves behind a drifting hole and therefore a positive charge in the semiconductor. Secondary (*i.e.*, additional electrons) then flow into the photoconductor to maintain neutrality in the sample and the current continues to flow. These events will continue until the hole has disappeared by recombination, which takes on average a time τ. Thus, more charges flow through the contact per unit time than charges actually photogenerated per unit time. What will happen if the contacts are not ohmic, *i.e.,* they are not injecting?

(c) What can you say about the product $\Delta\sigma$ and the speed of response, which is proportional to $1/\tau$.

5.22 Photoconductive detectors: PbSe for 4 μm detection The basic principle in photoconductive detectors is the change in the resistance of the semiconductor upon exposure to light. Figure 5.52 shows how PbS and PbSe photodetectors are often used. The photoconductor (PC) has a dark resistance R_d and is biased by V_B through a load R_L. A chopper (either mechanical or electronic) chops the light at a frequency f_c. The resistance of the PC changes periodically at the chopper frequency. The periodic change in R_d causes a periodic change in the current. This periodic change is the photocurrent $i_{ph}(t)$ signal and is an AC-type signal at the frequency f_c. The photocurrent generates a periodic voltage signal $v_{ph}(t)$ across R_d, which can be coupled through a coupling capacitor C_c into a lock-in amplifier (LIA). This amplifier is synchronized with the chopper and only amplifies the signal if it is in phase with the chopper. Its output is a DC signal that represents the magnitude of v_{ph} that is in phase with the chopped light. Consider a PbSe photoconductor for detecting 4 μm radiation. The circuit in Figure 5.52 has been implemented with $V_B = 15\,V$, $R_d = 1\,M\Omega$, and $R_L = R_d$. A test experiment has shown that incident optical power of 50 nW on the detector's photosensitive area ($3 \times 3\ mm^2$) decreases R_d by 6.0 Ω. (Put differently, $\delta R_d = -6\ \Omega$ for incident radiation of 50 nW.) Find the signal voltage V_{signal}, *i.e.,* the change δV in the voltage across R_d, and the photosensitivity, defined as V_{signal}/P_o. (b) What is the photosensitivity if $R_L = 2R_d$ and $R_L = R_d/2$? (c) For the case $R_L = R_d$, find the NEP and hence the specific detectivity D^*. Assume that the only noise source is the thermal noise from R_d and R_L. The quoted D^* for this device is $2.5 \times 10^9\ cm\ Hz^{1/2}\ W^{-1}$.

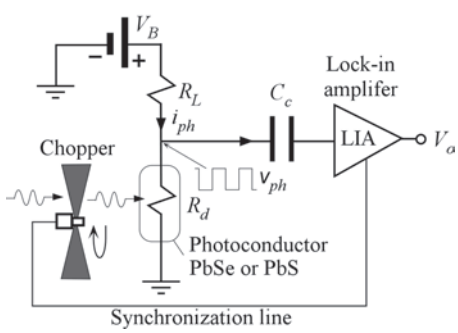


FIGURE 5.52 A photodetection system based on using a photoconductive detector, a chopper to chop the light, and a lock-in amplifier (LIA).

5.23 SNR of an InGaAs *pin* A particular photodetection application requires an InGaAs photodetector and needs a bandwidth of 2 GHz. The dark current of the InGaAs *pin* detector is 6 nA at 30°C. The minimum signal that is to be measured is 5 nW at 1550 nm where the responsivity is $0.95\ A\ W^{-1}$.

(a) Calculate the SNR in dB at 30°C.

(b) When the detector is cooled to −30°C, the dark current becomes 0.1 nA, and the responsivity is about the same. What is the new SNR?

(c) Suppose that we operate the detector over a bandwidth of 20 MHz. What is the new SNR at −30°C?

5.24 Detector and receiver noise Consider an InGaAs *pin* photodiode used in a receiver circuit as in Figure 5.31 with a load resistor of 30 kΩ. The total capacitance of the detector and the input of the amplifier together is 20 pF. The photodiode has a dark current of 1 nA. The incident radiation is 6 nW at 1550 nm where the responsivity is 0.9 A W^{-1}. Assuming that the amplifier is noiseless, calculate the SNR at 27°C.

5.25 The NEP and Ge and InGaAs photodiodes

(a) Show that the noise equivalent power of a photodiode is given by

$$\text{NEP} = \frac{P_1}{B^{1/2}} = \frac{hc}{\eta e \lambda}\left[2e(I_d + I_{ph})\right]^{1/2}$$

How would you improve the NEP of a photodiode? What is *NEP* for an ideal PD at operating at $\lambda = 1.55\,\mu$m?

(b) Given the dark current I_d of a PD, show that for SNR = 1, the photocurrent is

$$I_{ph} = eB\left[1 + \left(1 + \frac{2I_d}{eB}\right)^{1/2}\right]$$

What is the corresponding optical power P_1?

(c) Consider a fast Ge *pn* junction PD which has a photosensitive area of diameter 0.3 mm. It is reverse biased for photodetection and has a dark current of 0.5 μA. Its peak responsivity is 0.7 A V^{-1} at 1.55 μm (see Figure 5.47). The bandwidth of the photodetector and the amplifier circuit together is 100 MHz. Calculate its NEP at the peak wavelength and find the minimum optical power and hence minimum light intensity that gives a SNR of 1. How would you improve the minimum detectable optical power?

(d) Table 5.10 shows the characteristics of typical Ge *pn* junction and InGaAs *pin* photodiodes in terms of responsivity and the current. Fill in the remainder of the columns in the table assuming that there is an ideal, noiseless preamplifier to detect the photocurrent from the photodiode. Assume a working bandwidth, *B*, of 1 MHz. What is your conclusion?

TABLE 5.10 Ge *pn* junction and InGaAs *pin* PDs. Photosensitive area has a diameter of 1 mm

Photodiode	*R* at 1.55 μm (A V^{-1})	I_d (nA)	I_{ph} (nA) for SNR = 1 at B = 1 MHz	Optical power (nW) for SNR = 1 at B = 1 MHz	NEP (W $Hz^{-1/2}$)	Comment
Ge at 25°C	0.8	400				
Ge at –20°C	0.8	5				Thermoelectric cooling
InGaAs *pin*	0.95	3				

5.26 Specific detectivity of Si photodiodes Consider the NEP measurements shown in Table 5.11 for a collection of Si photodiodes. (a) For each calculate the specific detectivity. Then find the average and the mean standard deviation. (b) Plot NEP vs. $A^{1/2}$ on log-log plot and find the slope. What does the slope represent?

TABLE 5.11 Photosensitive area and NEP for a collection of Si photodiodes

A (mm^2)	7	13	20	25	66	100	324	784
NEP ($\times 10^{-15}$ W $Hz^{-1/2}$)	9.5	13	17	16	25	38	66	86

5.27 Specific detectivity of InGaAs photodiodes Consider the results of NEP measurements shown in Table 5.12 for a few InGaAs pin photodiodes with different photosensitive areas. (a) Plot the dark current vs. A on log-log plot and find the slope. What does the slope represent? (b) Plot NEP vs. A on log-log plot and find the slope. What does the slope mean?